Vision-Language-Action Models and World Models for Embodied AI: A Comprehensive Survey (2026)

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Abstract

The convergence of large-scale vision-language models, action-conditioned policy learning, and generative world modeling has catalyzed a paradigm shift in embodied artificial intelligence. Vision-Language-Action (VLA) models represent a new class of foundation models that directly map visual observations and natural language instructions to robot actions, enabling general-purpose manipulation, navigation, and interaction across diverse embodiments. Concurrently, world models—neural networks trained to predict future states of environments given current observations and actions—have emerged as powerful tools for planning, imagination-augmented reinforcement learning, and realistic environment simulation.

This survey provides a comprehensive and structured review of the theoretical foundations, architectural innovations, training methodologies, and application domains of VLA models and world models, with a particular focus on their synergistic integration for embodied AI. We begin by establishing the necessary background in transformer architectures, diffusion models, vision-language pre-training, and reinforcement learning. We then present a detailed taxonomy of VLA models, spanning from early robot transformer architectures such as RT-1 and RT-2 to modern flow-matching policies such as π_0 and $\pi_{0.5}$, open-source generalist policies like Octo and OpenVLA, and video-generation-augmented systems such as GR-2 and UniVLA. For world models, we trace the evolution from classical model-based reinforcement learning through recurrent world models (Ha and Schmidhuber, 2018), the Dreamer family (DreamerV1/V2/V3), to contemporary generative world simulators including UniSim, Genie, Genie 2, NVIDIA Cosmos, and diffusion-based game engines such as GameNGen and Oasis.

We systematically analyze how these two paradigms converge: VLA models increasingly leverage world model pre-training for robust visual representations and action prediction, while world models benefit from VLA-style action conditioning for controllable generation. We cover applications spanning real-world robotic manipulation (bin picking, dexterous grasping, mobile manipulation, autonomous driving), embodied navigation, and virtual world applications including game AI agents (VOYAGER, SPRING, STEVE-1), interactive game generation, and training environments for sim-to-real transfer. We further discuss benchmarks, evaluation metrics, datasets, and open challenges including scalability, safety, long-horizon reasoning, sim-to-real gaps, and the path toward unified foundation models that seamlessly integrate perception, prediction, planning, and action.

Our survey encompasses over 300 references from top-tier venues and influential technical reports, providing researchers and practitioners with a comprehensive roadmap for understanding and advancing the state of the art in VLA models, world models, and embodied artificial intelligence.

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1 Introduction

The quest to build intelligent agents that can perceive, reason, and act in the physical world has been a central aspiration of artificial intelligence since its inception [Turing, 1950, Newell and Simon, 1976]. Embodied AI—the study of agents that interact with and learn from environments through sensorimotor experience—has undergone a remarkable transformation in recent years, driven by the emergence of foundation models [Bommasani et al., 2021]. Two technological pillars have risen to prominence in this transformation: *Vision-Language-Action (VLA) models* and *world models*. Together, they promise to bridge the gap between abstract language understanding and concrete physical action, enabling machines that can understand natural language commands, predict the consequences of their actions, and execute complex tasks in both real and virtual environments.

1.1 The Rise of Foundation Models for Embodiment

The success of large language models (LLMs) such as GPT-4 [OpenAI, 2023a], PaLM [Chowdhery et al., 2023], and LLaMA [Touvron et al., 2023a], coupled with vision-language models (VLMs) like Flamingo [Alayrac et al., 2022], GPT-4V [OpenAI, 2023b], and LLaVA [Liu et al., 2024a], has demonstrated that pre-training on internet-scale data yields representations with remarkable generalization capabilities. A natural question arises: *Can the same pre-training paradigm be extended to produce general-purpose robot policies?* Vision-Language-Action models answer this question affirmatively by treating robot action prediction as a next-token prediction problem, leveraging the powerful visual and linguistic representations from pre-trained VLMs and fine-tuning on robot trajectory data [Brohan et al., 2023a, Kim et al., 2024, Black et al., 2024].

Simultaneously, the concept of a *world model*—a learned simulator of environment dynamics—has matured from theoretical ideas in model-based reinforcement learning [Sutton, 1991, Deisenroth et al., 2013] into practical systems capable of generating photorealistic video predictions [Yang et al., 2024, Bruce et al., 2024, OpenAI, 2024, NVIDIA, 2025]. World models serve dual purposes: they enable agents to "imagine" future outcomes for planning, and they provide synthetic training environments that can dramatically reduce the need for costly real-world data collection. The convergence of these two paradigms—action-conditioned policy learning and generative environment simulation—represents one of the most exciting frontiers in modern AI research.

1.2 Motivation and Scope

Despite rapid progress, the landscape of VLA models and world models for embodied AI remains fragmented across multiple research communities, including robotics, computer vision, natural language processing, reinforcement learning, and game AI. Existing surveys have covered subsets of this landscape: Wang et al. [2024a] surveyed vision-language-action models for manipulation, **?** reviewed world models for decision-making, and Lei et al. [2024] covered general-purpose agents for embodied AI. However, no existing work provides a *unified* treatment that:

1. Systematically covers the theoretical foundations of *both* VLA models and world models and their interconnections;
2. Traces the full evolution from classical model-based RL to modern generative world simulators;
3. Spans applications from real-world robotic manipulation to virtual game environments; and

4. Identifies the convergent trajectory toward unified perception-prediction-action foundation models.

This survey fills this gap. We provide a comprehensive, structured, and up-to-date review covering over 300 works from leading academic conferences (NeurIPS, ICML, ICLR, CVPR, RSS, CoRL, IROS, AAAI), preprint servers (arXiv), and influential industry technical reports (Google DeepMind, Physical Intelligence, NVIDIA, OpenAI, Toyota Research, Stanford, UC Berkeley, and others).

1.3 Contributions

The principal contributions of this survey are:

1. **Comprehensive Taxonomy:** We present a structured taxonomy of VLA models organized by architectural design (tokenized action spaces, continuous action regression, diffusion-based policies, flow-matching policies), training paradigm (co-training, fine-tuning, multi-task), and embodiment scope (single-robot, cross-embodiment, universal). For world models, we organize approaches by generative mechanism (autoregressive, diffusion-based, hybrid), temporal modeling strategy (recurrent state-space, frame-stacking, transformer), and application domain (RL planning, video generation, game simulation).

2. **Unified Theoretical Framework:** We present VLA models and world models within a common mathematical framework grounded in partially observable Markov decision processes (POMDPs), optimal transport theory for action generation, and variational inference for latent world state learning. This framework reveals the deep connections between action prediction and state prediction.

3. **Detailed System Analysis:** We provide in-depth analysis of landmark systems including RT-1/RT-2/RT-X [Brohan et al., 2023b,a, Collaboration et al., 2024], Octo [Ghosh et al., 2024], OpenVLA [Kim et al., 2024], $\pi_0/\pi_{0.5}$ [Black et al., 2024, Intelligence, 2025], DreamerV1/V2/V3 [Hafner et al., 2020, 2021, 2023], Genie/Genie 2 [Bruce et al., 2024, DeepMind, 2024], UniSim [Yang et al., 2024], GameNGen [Valevski et al., 2024], and NVIDIA Cosmos [NVIDIA, 2025], among many others.

4. **Cross-Domain Application Coverage:** We cover applications spanning real-world robotic manipulation (bin picking, assembly, dexterous grasping, mobile manipulation), autonomous driving, embodied navigation, surgical robotics, agricultural robotics, as well as virtual world applications including game AI agents (VOYAGER, SPRING, STEVE-1), interactive game generation, and sim-to-real transfer.

5. **Benchmarks and Evaluation:** We catalog the major benchmarks, datasets, and evaluation protocols used in VLA and world model research, including Open X-Embodiment, CALVIN, SIMPLER, LIBERO, RLBench, Habitat, and various game-based benchmarks.

6. **Open Challenges and Future Directions:** We identify and discuss the most pressing open challenges, including data scarcity for real-world robotics, the sim-to-real gap, long-horizon task composition, safety and robustness, computational efficiency, and the path toward truly general-purpose embodied foundation models.

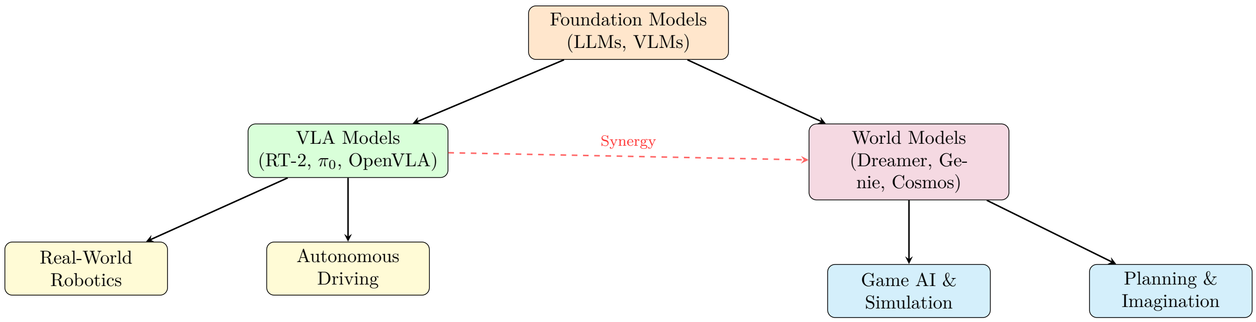

Figure 1: Overview of the survey structure. Foundation models (LLMs and VLMs) serve as the backbone for both VLA models and world models, which converge toward unified perception-prediction-action systems. Applications span real-world robotics, autonomous driving, game AI, and planning.

1.4 Organization of the Survey

The remainder of this survey is organized as follows. 2 establishes the necessary background in transformer architectures, diffusion models, vision-language pre-training, and reinforcement learning. 3 presents the theoretical foundations of VLA models. 4 provides a detailed taxonomy and analysis of VLA architectures and systems. 5 covers training paradigms, datasets, and scaling strategies. 6 discusses the theoretical foundations of world models. 7 reviews world model techniques and systems. 8 examines real-world embodied AI applications. 9 covers virtual world and game applications. 10 catalogs benchmarks and evaluation protocols. 11 discusses open challenges and future directions. 12 concludes the survey with a summary and outlook.

A visual overview of the survey structure and the relationship between VLA models, world models, and their applications is provided in 1.

1.5 Related Surveys and Positioning

To contextualize our contribution, we briefly review related surveys. Wang et al. [2024a] provided an early survey focused specifically on vision-language-action models for robotic manipulation, but their coverage predates several landmark systems including π_0, GR-2, and UniVLA. Du et al. [2024] surveyed LLM-based agents but focused primarily on text-based planning rather than embodied action. Lei et al. [2024] reviewed embodied AI from the perspective of general-purpose agents but with limited coverage of world models. Wang et al. [2024b] surveyed world models specifically for game environments. Karnan et al. [2024] provided a survey of world models for autonomous driving. In contrast, our survey provides the most comprehensive and unified treatment of both VLA models and world models, with systematic coverage of their theoretical connections, architectural innovations, training methodologies, and applications across both physical and virtual embodied AI.

2 Background and Preliminaries

In this section, we establish the foundational concepts and technical preliminaries that underpin both VLA models and world models. We begin with the formal framework of sequential decision-making, then review the core neural architectures, generative models, and pre-training paradigms that serve as building blocks for the systems discussed in subsequent sections.

2.1 Sequential Decision-Making Framework

Both VLA models and world models operate within the framework of sequential decision-making under uncertainty. We formalize this using the Partially Observable Markov Decision Process (POMDP) [Kaelbling et al., 1998], defined as a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \Omega, \gamma \rangle$ where:

- $\mathcal{S}$ is the state space (the true state of the environment, not directly observable);
- $\mathcal{A}$ is the action space (continuous, discrete, or hybrid);
- $\mathcal{O}$ is the observation space (e.g., images, language, proprioception);
- $T : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ is the transition dynamics;
- $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function;
- $\Omega : \mathcal{S} \to \Delta(\mathcal{O})$ is the observation function; and
- $\gamma \in [0, 1)$ is the discount factor.

A *policy* $\pi : \mathcal{O}^* \times \mathcal{L} \to \Delta(\mathcal{A})$ maps a history of observations and a language instruction $\ell \in \mathcal{L}$ to a distribution over actions. The goal of VLA models is to learn such a policy from demonstrations (imitation learning) or interaction (reinforcement learning). The goal of world models is to learn an approximate dynamics model $\hat{T} : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$ (or, more commonly, $\hat{T} : \mathcal{O} \times \mathcal{A} \to \Delta(\mathcal{O})$ operating directly in observation space) that can be used for planning, data augmentation, or simulation.

2.1.1 Imitation Learning

Most VLA models are trained via imitation learning from expert demonstrations. Given a dataset $\mathcal{D} = \{(o_t, \ell, a_t)\}_{t=1}^{N}$ of observation–instruction–action tuples, behavioral cloning seeks to minimize:

$$\mathcal{L}_{BC}(\theta) = \mathbb{E}_{(o,\ell,a)\sim\mathcal{D}} \left[-\log \pi_\theta(a \mid o, \ell) \right] \tag{1}$$

where π_θ is the parameterized policy. While simple, behavioral cloning suffers from distribution shift—compounding errors that arise when the agent visits states not seen during training [Ross et al., 2011]. Modern VLA models mitigate this through large-scale data diversity [Collaboration et al., 2024], action chunking [Zhao et al., 2023a], and diffusion-based action generation [Chi et al., 2023].

2.1.2 Reinforcement Learning

In settings where reward signals are available, policies can be optimized via reinforcement learning:

$$\max_\theta \; J(\theta) = \mathbb{E}_{\pi_\theta} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \tag{2}$$

Model-based RL methods learn a world model $\hat{T}_\phi$ and use it for planning [Finn and Levine, 2017, Chua et al., 2018, Janner et al., 2020] or synthetic data generation [Hafner et al., 2023]. Model-free methods such as PPO [Schulman et al., 2017] and SAC [Haarnoja et al., 2018] optimize the policy directly from environment interaction.

2.2 Transformer Architectures

The transformer architecture [Vaswani et al., 2017] has become the backbone of virtually all modern VLA and world model systems. We review its key components.

2.2.1 Self-Attention Mechanism

The core operation in transformers is scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

where $Q = XW_Q$, $K = XW_K$, $V = XW_V$ are the query, key, and value projections of the input sequence X, and d_k is the key dimension. Multi-head attention applies this operation in parallel across multiple heads, enabling the model to attend to different types of relationships simultaneously.

2.2.2 Vision Transformers

The Vision Transformer (ViT) [Dosovitskiy et al., 2021] adapts the transformer to image inputs by splitting an image into fixed-size patches, linearly embedding each patch, and processing the resulting sequence with standard transformer layers. Given an image $I \in \mathbb{R}^{H\times W\times C}$, ViT divides it into $N = HW/P^2$ patches of size $P \times P$, producing tokens $\{x_p^i\}_{i=1}^N$ that are projected and processed:

$$z_0 = [x_{\text{cls}}; x_p^1 E; x_p^2 E; \ldots ; x_p^N E] + E_{\text{pos}} \tag{4}$$

where E is the patch embedding matrix and E_{pos} encodes positional information. Variants such as SigLIP [Zhai et al., 2023], used in many VLA models, employ sigmoid loss for contrastive pre-training of the vision encoder.

2.2.3 Transformer Decoders and Encoder-Decoder Architectures

Many VLA models build on the encoder-decoder transformer architecture, originally proposed for sequence-to-sequence tasks [Vaswani et al., 2017]. The encoder processes the multimodal input (vision tokens, language tokens), while the decoder generates action tokens autoregressively. Notable examples include PaLM-E [Driess et al., 2023], which uses a decoder-only architecture, and RT-2 [Brohan et al., 2023a], which repurposes the VLM decoder for action token generation. The choice between decoder-only and encoder-decoder architectures has significant implications for inference speed and action generation quality, as we discuss in 4.

2.3 Diffusion Models

Diffusion models [Sohl-Dickstein et al., 2015, Ho et al., 2020, Song et al., 2021a] have emerged as the dominant paradigm for generating high-dimensional continuous outputs, including images, videos, and—importantly for VLA—action trajectories.

2.3.1 Denoising Diffusion Probabilistic Models (DDPMs)

DDPMs [Ho et al., 2020] define a forward process that gradually adds Gaussian noise to data:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{5}$$

where β_t is a noise schedule. The reverse process learns to denoise:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{6}$$

The training objective simplifies to:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t, c)\|^2\right] \tag{7}$$

where $\epsilon \sim \mathcal{N}(0, I)$ is the noise, $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon$ is the noised data, and c is an optional conditioning signal (e.g., visual observation and language instruction). The noise prediction network ϵ_θ is typically a U-Net [Ronneberger et al., 2015] or a transformer.

2.3.2 Diffusion for Action Generation

Chi et al. [2023] introduced Diffusion Policy, which applies the diffusion framework to generate robot action sequences conditioned on visual observations. Given an observation o_t and language instruction ℓ, the model generates an action chunk $\mathbf{a}_{t:t+H} = (a_t, a_{t+1}, \ldots, a_{t+H})$ by iteratively denoising from pure noise:

$$\mathbf{a}^{(k-1)}_{t:t+H} = \alpha\left(\mathbf{a}^{(k)}_{t:t+H} - \gamma\epsilon_\theta(\mathbf{a}^{(k)}_{t:t+H}, k, o_t, \ell)\right) + \sigma\epsilon' \tag{8}$$

This formulation naturally handles multimodal action distributions (e.g., multiple valid grasps for an object), which is a key advantage over deterministic policy representations.

2.3.3 Flow Matching

An alternative to diffusion that has gained traction in VLA models is flow matching [Lipman et al., 2023, Liu et al., 2023a], which learns a vector field that transports a simple prior distribution (e.g., Gaussian) to the target action distribution via an ordinary differential equation (ODE):

$$\frac{d\mathbf{a}}{dt} = v_\theta(\mathbf{a}, t, o, \ell) \tag{9}$$

Physical Intelligence's π_0 model [Black et al., 2024] pioneered the use of flow matching for VLA, demonstrating superior action generation quality with fewer denoising steps compared to diffusion.

2.4 Vision-Language Pre-training

Vision-Language Models (VLMs) form the representational backbone of VLA systems. We review the key pre-training paradigms.

2.4.1 Contrastive Pre-training

CLIP [Radford et al., 2021] and ALIGN [Jia et al., 2021] demonstrated that contrastive learning on large-scale image-text pairs produces transferable visual representations. Given an image–text pair (I, T), the contrastive loss maximizes the cosine similarity of matching pairs while minimizing that of non-matching pairs:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{\exp(\text{sim}(z_I^i, z_T^i)/\tau)}{\sum_{j=1}^{N}\exp(\text{sim}(z_I^i, z_T^j)/\tau)} \tag{10}$$

where z_I and z_T are the image and text embeddings, $\text{sim}(\cdot, \cdot)$ is cosine similarity, and τ is a learnable temperature parameter.

2.4.2 Generative Vision-Language Models

Models such as Flamingo [Alayrac et al., 2022], PaLI [Chen et al., 2023a], and LLaVA [Liu et al., 2024a] combine vision encoders with large language models through cross-attention or projection layers, enabling text generation conditioned on visual inputs. These architectures are particularly relevant for VLA models, as they provide a natural framework for processing visual observations and language instructions before mapping to actions.

2.4.3 Vision-Language-Action Pre-training

The key insight behind VLA models is extending vision-language pre-training to include action tokens. Rather than generating text, the model generates action representations. This can be done by:

1. **Tokenizing actions:** Discretizing continuous actions into tokens and treating action generation as a language modeling problem [Brohan et al., 2023a, **?**];
2. **Regression heads:** Adding continuous output heads on top of the vision-language backbone [Ghosh et al., 2024]; or
3. **Diffusion/Flow heads:** Attaching a diffusion or flow-matching head for action generation [Black et al., 2024, Chi et al., 2023].

2.5 Reinforcement Learning Preliminaries

We briefly review RL concepts relevant to world models.

2.5.1 Model-Based Reinforcement Learning

Model-based RL methods learn an approximate dynamics model $\hat{T}_\phi(s_{t+1} \mid s_t, a_t)$ and use it to improve the policy. The model can be used for:

- **Planning:** Searching for optimal action sequences via model-predictive control (MPC) [Finn and Levine, 2017, Chua et al., 2018];
- **Policy optimization:** Generating synthetic trajectories for policy training [Janner et al., 2020, Hafner et al., 2023]; or
- **Representation learning:** Learning useful state representations as a byproduct of dynamics prediction [Yarats et al., 2021].

2.5.2 Actor-Critic Methods

Modern model-based RL systems typically employ actor-critic architectures. The actor π_θ selects actions, while the critic $Q_\psi(s, a)$ or $V_\psi(s)$ estimates the value function. These are jointly optimized using objectives derived from temporal difference learning [Sutton and Barto, 2018] or advantage estimation [Schulman et al., 2016].

2.5.3 Sim-to-Real Transfer

A critical challenge for embodied AI is the gap between simulated training environments and the real world. Techniques for bridging this gap include domain randomization [Tobin et al., 2017], system identification [Yu et al., 2017], progressive nets [Rusu et al., 2016], and—most relevant to this survey—training world models that can generate realistic simulated experiences for policy learning [Yang et al., 2024].

2.6 Generative Video Models

The recent revolution in video generation models [Ho et al., 2022, Singer et al., 2023, Blattmann et al., 2023, OpenAI, 2024] has profoundly influenced both VLA and world model research. Video diffusion models learn to generate sequences of frames conditioned on text prompts, images, or actions. The core architecture typically involves:

- A 3D U-Net or transformer that operates on spatiotemporal volumes;
- Temporal attention layers that capture frame-to-frame dependencies;
- Conditioning mechanisms for text, images, or actions.

These models serve as the foundation for modern world models that generate future visual predictions conditioned on actions, enabling both planning and simulation as discussed in 7.

2.7 Summary

This section has established the key building blocks for the VLA and world model systems we survey in subsequent sections. 1 provides a summary of the key concepts. The interplay between transformer architectures, diffusion/flow-based generative models, vision-language pre-training, and model-based reinforcement learning forms the technical foundation upon which modern VLA models and world models are constructed.

3 VLA Theory and Foundations

In this section, we present the theoretical foundations of Vision-Language-Action (VLA) models. We formalize the VLA modeling problem, analyze different action representation strategies, discuss the expressiveness and limitations of various policy parameterizations, and examine the theoretical connections between VLA objectives and broader goals of embodied intelligence.

3.1 Problem Formulation

A VLA model seeks to learn a policy π_θ that maps a sequence of visual observations $o_{1:t}$, a natural language instruction ℓ, and optionally proprioceptive state s_t^{prop} to a distribution over actions:

$$\pi_\theta(a_t \mid o_{1:t}, \ell, s_t^{\text{prop}}) \tag{11}$$

The key distinction from standard vision-language models is that the output space $\mathcal{A}$ consists of executable actions (e.g., joint torques, end-effector displacements, navigation waypoints) rather than text tokens. This fundamental shift from symbolic to physical outputs introduces several theoretical challenges that we examine below.

Table 1: Summary of key technical building blocks for VLA and world models.

Component	**Role**	**Key References**
Transformers	Backbone architecture for encoding multimodal inputs and generating actions/states	[Vaswani et al., 2017, Dosovitskiy et al., 2021]
Diffusion Models	Generative modeling of continuous actions and future states	[Ho et al., 2020, Song et al., 2021a, Chi et al., 2023]
Flow Matching	Efficient continuous action generation via learned ODEs	[Lipman et al., 2023, Liu et al., 2023a, Black et al., 2024]
VLMs	Visual and linguistic representation learning	[Radford et al., 2021, Alayrac et al., 2022, Liu et al., 2024a]
Model-Based RL	Learning and leveraging environment dynamics	[Sutton, 1991, Deisenroth et al., 2013]
Video Generation	Spatiotemporal prediction for world simulation	[Ho et al., 2022, OpenAI, 2024]

3.1.1 The Action Representation Problem

The action space $\mathcal{A}$ varies dramatically across embodiments: a 7-DOF robot arm has a different action space than a mobile manipulator, which differs from a dexterous hand, which differs from a navigation agent. The choice of action representation has profound implications for the VLA model's ability to generalize across embodiments and tasks. We identify four principal action representation strategies:

Discretized Action Tokens. The action space is discretized into bins (e.g., 256 bins per dimension), and each bin is mapped to a token in the model's vocabulary. This approach, used by RT-1 [Brohan et al., 2023b], RT-2 [Brohan et al., 2023a], and OpenVLA [Kim et al., 2024], treats action prediction as a next-token prediction problem:

$$a_t = \text{DeTokenizer}(\text{argmax}_w \; P_\theta(w \mid o_{1:t}, \ell, w_{<a})) \tag{12}$$

where $w_{<a}$ denotes previously generated action tokens. The advantage is architectural simplicity—the same model that generates text can generate actions. The disadvantage is quantization error and difficulty representing multimodal action distributions.

Continuous Regression. The policy outputs continuous action values directly through a regression head. Octo [Ghosh et al., 2024] uses a diffusion-based head, while earlier approaches like BC-Z [Jang et al., 2022] used simple MLP heads:

$$a_t = f_\theta^{\text{reg}}(h_t) \tag{13}$$

where h_t is the latent representation from the VLA backbone. This avoids quantization but is limited to unimodal distributions without additional mechanisms.

Diffusion-Based Action Generation. Diffusion Policy [Chi et al., 2023] and its variants model the full conditional distribution over action sequences:

$$p_\theta(\mathbf{a}_{t:t+H} \mid o_{1:t}, \ell) = \int p(\mathbf{a}_{t:t+H}^{(0)}) \prod_{k=1}^{K} p_\theta(\mathbf{a}_{t:t+H}^{(k-1)} \mid \mathbf{a}_{t:t+H}^{(k)}, o_{1:t}, \ell) \, d\mathbf{a}^{(K:0)} \tag{14}$$

This naturally handles multimodal distributions and has been shown to outperform deterministic baselines on tasks with multiple valid strategies [Chi et al., 2023].

Flow Matching for Actions. π_0 [Black et al., 2024] uses flow matching to generate actions:

$$\mathbf{a}_1 = \mathbf{a}_0 + \int_0^1 v_\theta(\mathbf{a}_t, t, o, \ell) \, dt \tag{15}$$

Starting from a Gaussian prior $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, the learned vector field v_θ transports samples to the action distribution. Flow matching offers faster inference than diffusion (typically 4–10 integration steps vs. 50–100 denoising steps) while maintaining comparable distribution quality.

3.2 Expressiveness and Universal Approximation

A natural theoretical question is: *What class of policies can VLA models represent?* We address this for each action representation.

3.2.1 Tokenized Actions

When actions are discretized into V bins per dimension over d dimensions, the effective action vocabulary size is V^d for a single timestep, and $(V^d)^H$ for an action chunk of length H. For a

7-DOF robot with 256 bins and $H = 10$, this gives 256^{70} possible action sequences—far exceeding any practical vocabulary. In practice, models tokenize each dimension independently, yielding a sequence of $d \times H$ tokens, which is tractable but sacrifices inter-dimensional correlations in the early tokens.

Shridhar et al. [2023] showed that the tokenized action representation can approximate any continuous policy arbitrarily well as the number of bins increases, provided the model has sufficient capacity. However, the sample complexity of learning fine-grained tokenized policies grows rapidly with discretization resolution.

3.2.2 Diffusion and Flow-Based Policies

Diffusion and flow-matching policies are universal distribution approximators. Chen et al. [2023b] proved that score-based generative models can approximate any data distribution in total variation distance, given sufficient model capacity and data. The key theoretical advantage for action generation is the ability to represent *multimodal* conditional distributions $p(a \mid o, \ell)$—a critical capability for tasks like grasping, where multiple grasps may be equally valid.

3.3 Action Chunking and Temporal Consistency

A significant practical innovation in VLA models is *action chunking* [Zhao et al., 2023a], where the policy predicts a sequence of H future actions at each timestep rather than a single action:

$$\pi_\theta(\mathbf{a}_{t:t+H} \mid o_{1:t}, \ell), \quad \mathbf{a}_{t:t+H} = (a_t, a_{t+1}, \ldots, a_{t+H}) \tag{16}$$

Action chunking provides several theoretical and practical benefits:

1. **Reduced compounding error:** By committing to a planned sequence of actions, the policy avoids the cascading errors that arise from single-step predictions in behavioral cloning [Ross et al., 2011].

2. **Temporal smoothness:** Predicting action sequences naturally encodes the smoothness prior that most manipulation tasks require.

3. **Efficient execution:** The policy can be queried less frequently (once every H steps), reducing computational latency.

4. **Expressiveness for multi-step primitives:** Tasks like "pick up the cup" require coordinated multi-step motions that are better captured by action chunks than individual actions.

ACT [Zhao et al., 2023a] uses a conditional VAE (CVAE) to model the distribution over action chunks, with the CVAE encoder processing the ground-truth action sequence during training and the prior being sampled during inference. Diffusion Policy [Chi et al., 2023] replaces the CVAE with a diffusion process, achieving better multimodal action coverage.

3.4 Multimodal Fusion Strategies

A VLA model must fuse information from three modalities: vision (images or video), language (natural language instructions), and proprioception (robot state). We categorize the fusion strategies used in the literature.

3.4.1 Early Fusion

In early fusion, all modalities are tokenized and concatenated before being processed by a shared transformer. RT-2 [Brohan et al., 2023a] and OpenVLA [Kim et al., 2024] adopt this approach, interleaving visual tokens (from ViT), language tokens (from the LLM tokenizer), and action tokens in a single sequence:

$$\text{Input} = [\text{CLS}]; [v_1, \ldots, v_M]; [\ell_1, \ldots, \ell_L]; [a_1, \ldots, a_D] \tag{17}$$

This allows full bidirectional attention between all modalities but is computationally expensive for long sequences.

3.4.2 Late Fusion

In late fusion, each modality is processed by a separate encoder, and their representations are combined before action generation. Ghosh et al. [2024] use separate tokenizers for visual observations and actions, combining them in a shared transformer trunk. This is more modular and allows pre-trained encoders to be used for individual modalities.

3.4.3 Cross-Attention Fusion

An intermediate approach uses cross-attention layers to allow one modality to attend to another. PaLM-E [Driess et al., 2023] injects visual representations into a pre-trained LLM through cross-attention, preserving the LLM's text generation capability while grounding it in visual inputs. RoboFlamingo [Li et al., 2024a] adapts the Flamingo architecture [Alayrac et al., 2022] for robotic control using cross-attention between vision and language streams.

3.4.4 Embodiment-Aware Fusion

For cross-embodiment generalization, the fusion strategy must account for different robot morphologies. Octo [Ghosh et al., 2024] introduces embodiment-specific tokenizers that map each robot's observation and action spaces to a shared latent space. HPT [Allshire et al., 2024] proposes heterogeneous pre-training transformers that align different sensorimotor spaces through a shared representation. This enables a single policy to control diverse robots with different numbers of joints, sensor configurations, and action dimensions.

3.5 Generalization and Transfer Learning Theory

A central promise of VLA models is *generalization*—the ability to perform new tasks, handle new objects, and operate in new environments without task-specific training. We analyze the theoretical foundations of generalization in VLA models.

3.5.1 Pre-training and Transfer

The transfer learning theory of VLA models can be understood through the lens of representation learning. A VLM pre-trained on internet-scale data learns representations $\phi(x)$ of images and text that capture semantic structure. When fine-tuned on robot data, the model adapts these representations for action prediction. The key theoretical question is: *What properties of the pre-trained representation enable efficient transfer to robotic control?*

Brohan et al. [2023a] provided empirical evidence that VLM pre-training enables: (1) *semantic generalization*—understanding novel objects described in language; (2) *compositional generalization*—following novel instruction combinations (e.g., "pick up the *red* cup and place it in the *blue* bowl" when only individual concepts were seen during robot training); and (3) *chain-of-thought reasoning*—breaking complex instructions into executable steps.

3.5.2 Multi-Task and Cross-Embodiment Generalization

Training on data from multiple tasks and robots enables cross-task and cross-embodiment transfer. The Open X-Embodiment (OXE) dataset [Collaboration et al., 2024] aggregates data from 22 different robots across 13 institutions, demonstrating that RT-X models trained on this diverse data outperform single-robot baselines on held-out tasks.

From a theoretical perspective, multi-task learning can be understood as learning a shared representation that captures the common structure across tasks while allowing task-specific adaptation. The Rademacher complexity of the shared hypothesis class provides generalization bounds for multi-task VLA models [Baxter, 2000]. In practice, the diversity of the training data—in terms of tasks, objects, environments, and embodiments—is the primary driver of generalization performance.

3.5.3 Scale Laws for VLA Models

An emerging area of research concerns scaling laws for VLA models. Kim et al. [2024] showed that increasing model size from 3B to 7B parameters improves performance on held-out manipulation tasks, consistent with the scaling laws observed in language modeling [Kaplan et al., 2020]. However, the relationship between model scale, data scale, and task performance in VLA models is more complex than in pure language models due to the diversity of embodiments and the continuous nature of actions. The π_0 model [Black et al., 2024] demonstrated that training on the largest known robot dataset (from multiple robot types) with a 5B-parameter VLM backbone yields a generalist policy with remarkable zero-shot transfer capabilities, suggesting that VLA models benefit from scaling along both the data and model axes.

3.6 Connections to World Models

VLA models and world models are deeply connected at a theoretical level. A VLA model $\pi(a \mid o, \ell)$ can be decomposed into a world model component that predicts future observations given actions, and a planning component that selects actions to achieve the instruction:

$$\pi(a_t \mid o_{1:t}, \ell) = \arg\max_{a_t} \mathbb{E}_{\hat{o}_{t+1:T} \sim \hat{T}(\cdot \mid o_t, a_{t:T})} \left[R(\hat{o}_{t+1:T}, \ell) \right] \tag{18}$$

where $\hat{T}$ is the world model and R measures alignment with the instruction. In practice, most VLA models do not explicitly decompose in this way—they learn a direct mapping from observations to actions. However, the trend toward incorporating world model components (e.g., video prediction pre-training in GR-2 [Bao et al., 2025], action-conditioned video generation in UniVLA [Li et al., 2025]) suggests convergence toward systems that jointly learn to predict and act.

3.7 Safety and Robustness Considerations

Deploying VLA models in the real world raises critical safety concerns. Theoretical frameworks for safe VLA deployment include:

- **Constrained policy optimization:** Ensuring that actions satisfy safety constraints during training [Achiam et al., 2017];
- **Uncertainty estimation:** Detecting out-of-distribution situations where the model may produce unreliable actions [Chang et al., 2024];
- **Action bounds and filtering:** Constraining the action space to prevent dangerous motions [Chi et al., 2023];
- **Recovery behaviors:** Learning fallback actions when the primary policy fails [Fu et al., 2024].

The challenge is that VLA models inherit the failure modes of their underlying VLMs (hallucination, misinterpretation of instructions) with potentially physical consequences. Developing robust safety mechanisms for VLA models remains an active area of research.

3.8 Summary

In this section, we have established the theoretical foundations of VLA models, covering action representations (tokenized, continuous, diffusion-based, flow-matched), multimodal fusion strategies, generalization theory, and connections to world models. The key theoretical insight is that VLA models extend the successful pre-training paradigm from language and vision to physical action, but face unique challenges arising from the continuous, multimodal, and safety-critical nature of embodied control. In the next section, we examine how these theoretical foundations are realized in concrete architectural designs and systems.

4 VLA Architectures and Systems

In this section, we provide a detailed taxonomy and analysis of VLA model architectures and systems. We organize our discussion along several dimensions: the base VLM architecture, the action generation mechanism, the scope of embodiment generalization, and key design innovations. We cover the most influential systems in the literature, from the pioneering RT-1 to the state-of-the-art $\pi_{0.5}$ and beyond.

4.1 Taxonomy of VLA Architectures

2 presents our taxonomy of VLA architectures. We identify three primary design axes:

1. **Base Architecture:** Decoder-only (e.g., RT-2, OpenVLA), encoder-decoder (e.g., Octo), or modular (e.g., π_0, Diffusion Policy).
2. **Action Generation:** Autoregressive token prediction (RT-1/RT-2/OpenVLA), diffusion-based (Diffusion Policy, RDPS), or flow-matching (π_0/$\pi_{0.5}$).
3. **Embodiment Scope:** Single robot (RT-1), multi-task (Diffusion Policy, ACT), cross-embodiment (Octo, RT-X, HPT), or universal (π_0, $\pi_{0.5}$).

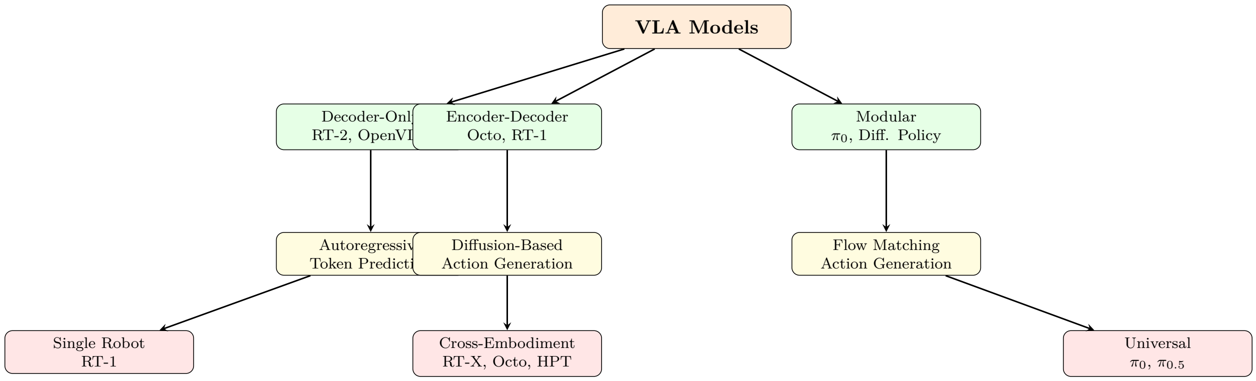

Figure 2: Taxonomy of VLA model architectures organized along three axes: base architecture, action generation mechanism, and embodiment scope.

4.2 The RT Family: Robot Transformers

4.2.1 RT-1: Robot Transformer 1

RT-1 [Brohan et al., 2023b] was the first large-scale transformer-based policy for robotic manipulation. It was trained on 130k episodes of demonstration data collected across 13 robots and over 700 tasks in a kitchen environment. The architecture processes images from a robot-mounted camera through a pre-trained EfficientNet [Tan and Le, 2019] backbone, tokenizes the visual features with a Token Learner [Ryoo et al., 2021], and feeds them into a Transformer decoder that autoregressively generates discretized action tokens.

The action space in RT-1 consists of 7 dimensions: 3 for position displacement $(\Delta x, \Delta y, \Delta z)$, 3 for rotation (roll, pitch, yaw), and 1 for gripper closure. Each dimension is discretized into 256 bins, resulting in $7 \times 8 = 56$ bits per action. The model generates these as 8 tokens of a vocabulary of 256 using a standard autoregressive decoder.

Key innovations of RT-1 include:

- Demonstrating that large-scale transformer-based policies can achieve strong performance on diverse manipulation tasks;
- Introducing the Token Learner for efficient visual tokenization;
- Showing that the model can incorporate language instructions at inference time for zero-shot task specification.

RT-1 achieved a 97% success rate on the training task distribution and showed preliminary evidence of generalization to new objects and instructions.

4.2.2 RT-2: Vision-Language-Action Models

RT-2 [Brohan et al., 2023a] represents a paradigm shift by directly fine-tuning a pre-trained vision-language model for robot control. Rather than training a policy from scratch, RT-2 builds on PaLI-X [Chen et al., 2023a] (55B parameters) and PaLM-E [Driess et al., 2023] (12B parameters), treating robot actions as just another language to be generated. The key insight is that VLMs already understand visual concepts and language semantics—they only need to learn the "language" of robot actions.

The fine-tuning process co-trains on internet-scale vision-language data and robot demonstration data. Actions are tokenized using the same scheme as RT-1 and appended to the model's vocabulary. During inference, the model receives a visual observation and language instruction, and generates action tokens using standard autoregressive decoding.

RT-2's major contributions include:

- **Emergent generalization:** The model can follow instructions involving objects and concepts never seen in robot training data, leveraging knowledge from VLM pre-training. For example, it can pick up an "extinct animal" (a dinosaur toy) despite never having seen one during robot training.
- **Chain-of-thought reasoning:** When prompted with multi-step reasoning instructions, RT-2 can decompose complex commands into executable action sequences.
- **Minimal robot data:** Despite being a 55B-parameter model, RT-2 requires robot data representing only a small fraction of its total training compute.

RT-2 achieved 3x improvement over RT-1 on tasks requiring semantic generalization and 2x improvement on tasks requiring multi-step reasoning, while maintaining comparable performance on the original training tasks.

4.2.3 RT-X and Open X-Embodiment

RT-X [Collaboration et al., 2024] extends the RT family to cross-embodiment learning. The Open X-Embodiment (OXE) dataset aggregates data from 22 different robots across 13 institutions, totaling over 1 million episodes spanning 500+ skills. RT-X models are trained on this diverse dataset, demonstrating that a single policy can control multiple robot platforms.

The OXE dataset standardizes the data format across robots, mapping diverse observation and action spaces to a common schema. Key findings from the RT-X experiments include:

- Positive transfer: training on data from multiple robots improves performance on each individual robot, even when the robots have different morphologies.
- Scaling with data diversity: performance improves as more diverse robot data is added, suggesting that cross-embodiment training is complementary to single-robot training.
- The "robot internet": analogous to how internet-scale data enabled general language models, diverse robot data enables more general robot policies.

4.3 Octo: Open-Source Generalist Robot Policy

Octo [Ghosh et al., 2024] is an open-source, generalist robot policy designed for cross-embodiment adaptation. Built on a transformer encoder-decoder architecture, Octo introduces several key design choices:

1. **Blockwise attention:** The transformer processes observation tokens, action tokens, and readout tokens in separate blocks with configurable attention patterns, enabling efficient processing of diverse input modalities.
2. **Embodiment-specific tokenizers:** Each robot platform has its own observation tokenizer that maps raw sensory inputs to a shared latent space, and its own action head that decodes latent representations to robot-specific actions.

3. **Diffusion action head:** Rather than autoregressive token generation, Octo uses a diffusion-based action head that generates action chunks through iterative denoising, naturally handling multimodal action distributions.

4. **Fine-tuning via readout tokens:** Novel tasks and embodiments can be adapted by adding new readout tokens and fine-tuning the action head, without modifying the shared transformer trunk.

Octo was trained on the OXE dataset comprising 800k episodes from 4 different robots. It demonstrates strong zero-shot and few-shot transfer to new robots and tasks, establishing a strong open-source baseline for generalist robot policies. The model is available in 27M and 93M parameter variants, balancing capability and deployability.

4.4 OpenVLA: Open-Source Vision-Language-Action Model

OpenVLA [Kim et al., 2024] is an open-source 7B-parameter VLA model built on Llama 2 [Touvron et al., 2023b] and a SigLIP [Zhai et al., 2023] vision encoder. Following the RT-2 paradigm, OpenVLA fine-tunes a pre-trained VLM on robot demonstration data, treating action tokens as an extension of the language vocabulary.

Key features of OpenVLA include:

- **Efficient fine-tuning:** Low-rank adaptation (LoRA) [Hu et al., 2022a] enables fine-tuning the 7B model on consumer GPUs, democratizing access to large-scale VLA models.
- **Cross-embodiment evaluation:** OpenVLA was evaluated on WidowX Bridge V2 data, achieving state-of-the-art performance on manipulation tasks.
- **Compositional generalization:** The model can follow novel instruction compositions, leveraging the linguistic knowledge from LLM pre-training.

OpenVLA represents an important milestone in making VLA research accessible, providing a fully open-source pipeline from data processing to model training to deployment.

4.5 Physical Intelligence: π_0 and $\pi_{0.5}$

4.5.1 π_0: A Generalist VLA Policy via Flow Matching

π_0 [Black et al., 2024], developed by Physical Intelligence, represents a significant advancement in generalist VLA policies. The model is built on a VLM backbone (PaliGemma-style architecture with a 400M ViT and 3B language model) and uses flow matching for action generation.

The key architectural innovations of π_0 are:

1. **Flow matching action head:** Rather than autoregressive token generation or diffusion, π_0 uses a flow matching head that generates continuous action trajectories via learned vector fields. This allows for fast action generation (4–10 integration steps) while maintaining high action quality.

2. **Largest robot training dataset:** π_0 was trained on a dataset comprising data from multiple robot platforms (single-arm, bimanual, mobile manipulator), totaling the largest known collection of robot demonstration data.

3. **Hierarchical architecture:** A vision-language module processes the multimodal input and produces a latent representation, which is then used to condition the flow matching action head.

π_0 demonstrates remarkable capabilities:

- Folding laundry, busing tables, and assembling boxes straight from training;
- Zero-shot transfer to novel objects and environments;
- Fine-tuning to complex multi-stage tasks like making coffee or cleaning kitchens with only tens to hundreds of demonstrations.

4.5.2 $\pi_{0.5}$: Towards Generalist Robot Intelligence

$\pi_{0.5}$ [Intelligence, 2025] extends π_0 with enhanced reasoning capabilities and broader task coverage. Key improvements include:

- Integration of chain-of-thought reasoning for decomposing complex instructions;
- Training on an even larger and more diverse dataset;
- Improved flow matching with adaptive step scheduling;
- Enhanced safety mechanisms for real-world deployment.

$\pi_{0.5}$ demonstrates the ability to perform household tasks such as making breakfast, organizing a kitchen, and loading a dishwasher, representing a significant step toward truly general-purpose robot intelligence.

4.6 GR-1 and GR-2: Video Prediction Augmented VLA

4.6.1 GR-1

GR-1 [Huang et al., 2024] introduces a novel pre-training strategy that leverages video prediction as a bridge between VLM pre-training and robot policy learning. The key idea is to pre-train a transformer model to predict future video frames given an observation and action sequence, and then fine-tune this model for robot control.

The architecture consists of:

1. A visual encoder (pre-trained ViT) that extracts tokens from image observations;
2. A transformer that processes the sequence of observation and action tokens;
3. A video prediction head used during pre-training; and
4. An action prediction head used during fine-tuning.

GR-1 is pre-trained on Ego4D [Grauman et al., 2022] and human manipulation videos, learning rich representations of physical dynamics. When fine-tuned on CALVIN [Mees et al., 2022] and real-world robot data, GR-1 outperforms baselines that do not use video pre-training, demonstrating the value of world model-style pre-training for VLA models.

4.6.2 GR-2

GR-2 [Bao et al., 2025] scales up the video prediction pre-training approach to unprecedented levels. Pre-trained on 38 million video clips from diverse sources, GR-2 demonstrates that massive video pre-training significantly improves VLA performance. The model generates future video frames conditioned on candidate actions, enabling "mental simulation" of action outcomes before execution.

Key results for GR-2 include:

- State-of-the-art performance on 18+ real-world manipulation tasks;
- Strong generalization to novel objects, backgrounds, and environments;
- Successful execution of long-horizon tasks requiring temporal reasoning.

4.7 Other Notable VLA Systems

4.7.1 Diffusion Policy

Diffusion Policy [Chi et al., 2023] applies the denoising diffusion framework to action generation, representing a paradigm shift from deterministic action regression. The model conditions on visual observations (processed by a ViT or CNN encoder) and generates action chunks through iterative denoising.

Two architectural variants are presented:

- **CNN-based:** A U-Net processes the noised action sequence conditioned on visual features, suitable for lower-dimensional action spaces.
- **Transformer-based:** A transformer processes the action sequence with cross-attention to visual features, better handling variable-length action chunks.

Diffusion Policy achieves state-of-the-art results on 12 out of 15 tasks in the Robomimic benchmark and all 4 tasks in the Block Pushing benchmark, establishing diffusion as a powerful action generation mechanism.

4.7.2 ACT: Action Chunking with Transformers

ACT [Zhao et al., 2023a] introduces action chunking (predicting sequences of future actions) and a conditional VAE to model the distribution over action chunks. The architecture uses a stereo camera setup and trains with a combination of behavioral cloning and the CVAE objective.

ACT was demonstrated on the ALOHA bimanual platform [Zhao et al., 2023a], achieving impressive performance on tasks requiring precise bimanual coordination such as sliding a zipper, inserting a battery, and threading a cable.

4.7.3 ALOHA and Mobile ALOHA

ALOHA [Zhao et al., 2023a] (A Low-cost Open-source Hardware System for Bimanual Teleoperation) provides an accessible hardware platform for collecting demonstration data. The system consists of two low-cost robot arms with leader-follower teleoperation, enabling rapid data collection. Mobile ALOHA [Fu et al., 2024] extends this with a mobile base, enabling whole-body manipulation tasks that combine locomotion and arm control.

The ALOHA system has been instrumental in democratizing VLA research, as the low cost (under $20,000) makes it accessible to academic labs. Data collected on ALOHA has been used to train various VLA models, including ACT and Diffusion Policy.

4.7.4 RoboFlamingo

RoboFlamingo [Li et al., 2024a] adapts the Flamingo [Alayrac et al., 2022] architecture for robotic control. It uses a pre-trained vision encoder, a language model, and cross-attention layers to fuse visual and linguistic information, followed by an MLP regression head for action prediction. RoboFlamingo demonstrates that VLM architectures designed for visual question answering can be repurposed for robotic control with minimal architectural changes.

4.7.5 TraceVLA

TraceVLA [Xiao et al., 2024] introduces visual trace prompts that overlay task-relevant visual markers (e.g., bounding boxes, arrows) on the input images. These traces provide additional spatial reasoning capabilities to the VLA model, improving performance on tasks requiring precise spatial understanding. TraceVLA fine-tunes a pre-trained VLA model with a mixture of trace-augmented and standard data, demonstrating improved spatial reasoning without sacrificing general language understanding.

4.7.6 UniVLA

UniVLA [Li et al., 2025] unifies video generation and action prediction within a single model. The model generates future video frames conditioned on actions (world model capability) and generates actions conditioned on observations and instructions (VLA capability), sharing the same backbone. This joint training enables mutual reinforcement between the two objectives: video prediction provides temporal reasoning for actions, and action conditioning improves the physical consistency of video generation.

4.7.7 HPT: Heterogeneous Pre-training Transformers

HPT [Allshire et al., 2024] addresses the challenge of training a single policy across robots with different sensorimotor spaces. The key innovation is a pair of "alignment" modules: an input alignment module that maps heterogeneous sensory inputs to a shared representation, and an output alignment module that maps the shared representation to heterogeneous action spaces. This enables pre-training on diverse robot data without requiring a standardized observation or action format.

4.7.8 GAP: Generalist Agent Policy

GAP [Nair et al., 2024] proposes a simple yet effective architecture for cross-embodiment generalization. The model uses a single transformer backbone with lightweight modality-specific adapters, processing visual observations, language instructions, and proprioceptive inputs. GAP demonstrates that with sufficient training data diversity, even relatively simple architectures can achieve strong cross-embodiment transfer.

Table 2: Comparison of major VLA model architectures. "AR" denotes autoregressive, "Diff." denotes diffusion, "FM" denotes flow matching, "CVAE" denotes conditional variational autoencoder, and "Reg." denotes regression.

Model	Action Rep.	Base Arch.	Params	Training Data	Embodiment Scope	Key Innovation
RT-1 [Brohan et al., 2023b]	AR Tokens	Enc-Dec	35M	130k episodes	Single robot	First robot transformer
RT-2 [Brohan et al., 2023a]	AR Tokens	Decoder-only	55B	VLM + robot data	Single robot	VLM fine-tuning
RT-X [Collaboration et al., 2024]	AR Tokens	Both	55B	OXE (1M+ episodes)	Cross-embodiment	Open X-Embodiment
Octo [Ghosh et al., 2024]	Diff.	Enc-Dec	93M	OXE (800k episodes)	Cross-embodiment	Open-source generalist
OpenVLA [Kim et al., 2024]	AR Tokens	Decoder-only	7B	OXE + Bridge	Cross-embodiment	Open-source VLA
π_0 [Black et al., 2024]	FM	Modular	3B+	Proprietary large	Universal	Flow matching
$\pi_{0.5}$ [Intelligence, 2025]	FM	Modular	–	Proprietary larger	Universal	Reasoning + actions
GR-1 [Huang et al., 2024]	Reg.	Enc-Dec	–	Video + CALVIN	Single robot	Video pre-training
GR-2 [Bao et al., 2025]	Reg.	Enc-Dec	–	38M videos	Single robot	Massive video pre-training
Diff. Policy [Chi et al., 2023]	Diff.	CNN/Trans.	100M	Task-specific	Single robot	Diffusion for actions
ACT [Zhao et al., 2023a]	CVAE	Enc-Dec	–	Task-specific	Single robot	Action chunking
UniVLA [Li et al., 2025]	Diff./AR	Enc-Dec	–	Video + robot	Cross-embodiment	Unified VLA + WM
HPT [Allshire et al., 2024]	Reg.	Modular	–	Diverse robots	Cross-embodiment	Alignment modules

4.7.9 CogAct

CogAct [Li et al., 2024b] introduces a cognitive architecture that decomposes VLA into a high-level task planner (VLM-based) and a low-level action executor (diffusion policy). The planner generates language subgoals that the executor conditions on, enabling long-horizon task completion through hierarchical decomposition. This two-level architecture achieves state-of-the-art performance on long-horizon manipulation benchmarks.

4.8 Comparative Analysis

2 provides a comprehensive comparison of the major VLA systems discussed in this section, organized by architecture, action representation, model size, training data, and key capabilities.

4.9 Design Principles and Lessons Learned

Our analysis of VLA architectures reveals several key design principles:

1. **Pre-training is essential:** All high-performing VLA models leverage pre-trained visual and linguistic representations. The shift from training policies from scratch (RT-1) to fine-tuning pre-trained VLMs (RT-2, OpenVLA, π_0) has been transformative.

2. **Continuous action representations outperform discretized tokens for precision tasks:** While tokenized actions are simple and leverage the language modeling infrastructure, diffusion and flow-matching representations achieve superior performance on tasks requiring precise continuous control.

3. **Cross-embodiment training improves single-robot performance:** Training on diverse robot data (RT-X, Octo) consistently improves performance even on individual robots, suggesting that cross-embodiment learning captures transferable manipulation primitives.

4. **World model pre-training enhances VLA performance:** Systems that pre-train with video prediction (GR-1, GR-2) or integrate world model components (UniVLA) demonstrate improved temporal reasoning and physical understanding.

5. **Scale matters, but architecture matters too:** While larger models generally perform better, architectural innovations like flow matching (π_0) and action chunking (ACT) can yield significant improvements at smaller model sizes.

4.10 Summary

In this section, we have presented a comprehensive taxonomy and analysis of VLA architectures and systems. The field has progressed rapidly from the first robot transformers (RT-1) to generalist policies capable of controlling diverse robots (π_0, $\pi_{0.5}$), with key innovations in action representation, multimodal fusion, and pre-training strategies. The convergence with world model techniques (GR-2, UniVLA) is a particularly promising trend that we explore further in subsequent sections.

5 Training Paradigms for VLA Models

The training methodology is a critical determinant of VLA model performance. In this section, we examine the training paradigms, datasets, data collection strategies, and scaling approaches that underpin modern VLA systems. We also discuss the role of data quality, augmentation, and the emerging practice of training on "internet-scale" video data.

5.1 Training Objectives and Paradigms

5.1.1 Behavioral Cloning with Autoregressive Action Prediction

The simplest and most widely used training paradigm for VLA models is behavioral cloning with autoregressive action prediction. Given a dataset of expert demonstrations $\mathcal{D} = \{(o_t, \ell, \mathbf{a}_{t:t+H})\}$, the model is trained to maximize the log-likelihood of expert actions:

$$\mathcal{L}_{AR}(\theta) = -\mathbb{E}_{(o,\ell,\mathbf{a})\sim\mathcal{D}}\left[\sum_{i=1}^{D\times H} \log P_\theta(a_i \mid o, \ell, a_{<i})\right] \tag{19}$$

where D is the action dimensionality and H is the action chunk horizon. This paradigm is used by RT-1 [Brohan et al., 2023b], RT-2 [Brohan et al., 2023a], and OpenVLA [Kim et al., 2024].

A key design choice is whether to include the visual observation at every token position (full cross-attention) or only at the beginning (prefix attention). Full cross-attention allows the model to attend to different visual features when generating each action dimension but is more computationally expensive.

5.1.2 Co-training with Vision-Language Data

A breakthrough insight from RT-2 [Brohan et al., 2023a] is that co-training on robot demonstration data alongside internet-scale vision-language data yields better VLA models than training on robot data alone. The co-training objective combines the robot action prediction loss with the standard VLM loss:

$$\mathcal{L}_{\text{co-train}}(\theta) = \lambda_{VL}\mathcal{L}_{VL}(\theta) + \lambda_{\text{robot}}\mathcal{L}_{AR}(\theta) \tag{20}$$

where $\mathcal{L}_{VL}$ is the standard vision-language modeling loss (image captioning, VQA, etc.) and $\lambda_{VL}, \lambda_{\text{robot}}$ are balancing coefficients. The benefit is two-fold: (1) the VLM pre-training knowledge is not forgotten during robot fine-tuning (avoiding catastrophic forgetting), and (2) the robot data benefits from the rich visual-linguistic representations learned from internet data.

In practice, the co-training mixture typically consists of a small fraction of robot data (e.g., 1–10% of training tokens) combined with a large volume of VL data. This means that the model "sees" each robot demonstration many times, which is necessary given the relatively small size of robot datasets compared to internet text.

5.1.3 Diffusion-Based Training

For diffusion-based VLA models like Diffusion Policy [Chi et al., 2023] and Octo [Ghosh et al., 2024], the training objective is the standard denoising score matching loss:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{(\mathbf{a},o,\ell)\sim\mathcal{D},k,\epsilon}\left[\|\epsilon - \epsilon_\theta(\mathbf{a}^{(k)}, k, o, \ell)\|^2\right] \tag{21}$$

where $k \sim \text{Uniform}(1, K)$ is the diffusion timestep and $\mathbf{a}^{(k)} = \sqrt{\bar{\alpha}_k}\mathbf{a} + \sqrt{1-\bar{\alpha}_k}\epsilon$ is the noised action sequence. The observation o and language instruction ℓ condition the denoising network through cross-attention or FiLM layers [Perez et al., 2018].

A practical consideration is the choice of the diffusion horizon K. While DDPM [Ho et al., 2020] used $K = 1000$, diffusion policies typically use $K = 100$ or fewer, and accelerated sampling methods like DDIM [Song et al., 2021b] can reduce inference to 10–20 denoising steps.

5.1.4 Flow Matching Training

For flow-matching-based VLA models like π_0 [Black et al., 2024], the training objective minimizes the expected flow matching loss:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,\mathbf{a}_0,\mathbf{a}_1}\left[\|v_\theta(\mathbf{a}_t, t, o, \ell) - (\mathbf{a}_1 - \mathbf{a}_0)\|^2\right] \tag{22}$$

where $\mathbf{a}_0 \sim \mathcal{N}(0, I)$ is a sample from the prior, $\mathbf{a}_1$ is the target action sequence from the dataset, $\mathbf{a}_t = (1-t)\mathbf{a}_0 + t\mathbf{a}_1$ is a linear interpolation, and v_θ is the vector field to be learned. This simpler training objective (no noise schedule to tune) and faster inference (fewer integration steps) make flow matching an attractive alternative to diffusion for VLA models.

5.1.5 Multi-Task and Multi-Embodiment Training

Training a single VLA model on data from multiple tasks and robots introduces additional design choices:

- **Task balancing:** Ensuring that the model learns equally well across tasks despite varying data quantities. Octo [Ghosh et al., 2024] uses a sampling strategy that oversamples underrepresented tasks.
- **Embodiment conditioning:** The model must know which robot it is controlling. This is typically achieved through a one-hot embodiment token or a learned embodiment embedding.
- **Shared vs. separate action heads:** Should all robots share the same action output head, or should each robot have its own? Octo uses separate heads, while RT-X uses a unified discretized action space.

5.2 Datasets for VLA Training

5.2.1 Robot Demonstration Datasets

The availability of large-scale robot demonstration datasets has been a key enabler of VLA research. 3 summarizes the major datasets.

Open X-Embodiment (OXE) [Collaboration et al., 2024] is the largest cross-embodiment robot dataset, aggregating data from 22 different robots across 13 institutions. It includes manipulation, navigation, and interaction data collected through teleoperation, scripted policies, and

Table 3: Major datasets used for VLA model training.

Dataset	Source	Episodes	Robots	Tasks
RT-1 Dataset	Google DeepMind	130k	13 (same type)	700+
Open X-Embodiment (OXE)	13 institutions	1M+	22	500+
Bridge Data V2	UC Berkeley	24k	1 (WidowX)	24
DROID	Multiple	76k	13	56
Bridge V2 + RT-1	Berkeley + Google	150k+	2	700+
CALVIN	NVIDIA	24h of play	Franka	34 tasks
RLBench	Imperial College	100 demos/task	Simulated	100 tasks
ALOHA Dataset	Stanford	Varies	ALOHA (bimanual)	Varies
OXO (Oxford)	Oxford	3M+ frames	Multiple	Manipulation

human demonstration. The dataset uses a standardized format that maps diverse observation and action spaces to a common schema.

Bridge Data V2 [Walke et al., 2023] provides 24k episodes of tabletop manipulation on a WidowX robot, covering 24 task categories. It is widely used as a benchmark for VLA model evaluation, particularly for testing generalization to new objects and instructions.

CALVIN [Mees et al., 2022] is a benchmark for long-horizon language-conditioned manipulation in a simulated tabletop environment. It provides 34 tasks organized by difficulty, with evaluation on composing sequences of up to 5 tasks.

DROID [Collaboration, 2024] is a large-scale, diverse robot manipulation dataset collected across 13 institutions using Franka robots. It includes 76k trajectories across 56 task categories with multiple camera views.

5.2.2 Vision-Language Pre-training Data

VLA models typically initialize from VLMs pre-trained on internet-scale data:

- **Image-text pairs:** Datasets like LAION-5B [Schuhmann et al., 2022], COCO Captions [Chen et al., 2015], and Conceptual Captions [Sharma et al., 2018] provide billions of image-text pairs for contrastive or generative pre-training.

- **Interleaved image-text data:** Datasets like OBELICS [Laurençon et al., 2024] provide interleaved image-text sequences that better match the format of VLA inputs.

- **Instruction tuning data:** Datasets like LLaVA-Instruct [Liu et al., 2024a] provide visual instruction-following data that aligns the model's outputs with human instructions.

5.2.3 Video Pre-training Data

An emerging trend is to pre-train on video data before fine-tuning on robot demonstrations. GR-2 [Bao et al., 2025] pre-trains on 38 million video clips from diverse sources, learning rich representations of physical dynamics and object interactions. This approach leverages the massive amount of video data available online (e.g., YouTube, Ego4D [Grauman et al., 2022]) to learn world dynamics that transfer to robotic control.

5.3 Data Collection Strategies

5.3.1 Teleoperation

The most common approach to collecting robot demonstrations is teleoperation, where a human operator controls the robot through a leader device, joystick, or VR controller. Systems like ALOHA [Zhao et al., 2023a] use a leader-follower setup where the operator moves a "leader" arm and the "follower" robot arm replicates the motion. This provides high-quality demonstrations but is time-consuming—a typical ALOHA session collects 50 demonstrations in about an hour.

5.3.2 Scripted Policies

For simulated environments, demonstrations can be generated by scripted policies (hard-coded controllers). RLBench [James et al., 2020] uses scripted policies to generate demonstrations for 100 tasks. While scalable, scripted policies may not exhibit the natural variability of human demonstrations.

5.3.3 Human Video Data

An emerging approach is to train VLA models on human video data of manipulation (e.g., Ego4D [Grauman et al., 2022], Something-Something [Goyal et al., 2017]). The challenge is extracting action labels from video—human hands and robot manipulators have different kinematics. Recent work on hand-to-robot retargeting [Bharadhwaj et al., 2024] and inverse kinematics estimation from video [Bahl et al., 2022] aims to bridge this gap.

5.3.4 Autonomous Data Collection

Several works explore autonomous data collection strategies where the robot generates its own training data through exploration. These include curiosity-driven exploration [Pathak et al., 2017], goal-conditioned data collection [Pong et al., 2022], and self-improving systems that iteratively collect data, train, and evaluate [Pinto and Gupta, 2016].

5.4 Data Augmentation and Processing

5.4.1 Visual Augmentation

Common visual augmentations for VLA training include:

- Random cropping and resizing;
- Color jittering (brightness, contrast, saturation, hue);
- Random erasing to simulate occlusions;
- Gaussian noise injection;
- Perspective transformations.

These augmentations help the model generalize to visual variations encountered in the real world (lighting changes, camera pose variations, occlusions).

5.4.2 Language Augmentation

To improve robustness to instruction phrasing, several approaches augment the language instructions:

- Paraphrasing: Generating alternative phrasings of the same instruction using LLMs [Brohan et al., 2023a];
- Template-based variation: Using templates with interchangeable synonyms (e.g., "pick up the {cup, mug, glass}");
- Negative instructions: Including instructions that describe what *not* to do.

5.4.3 Action Augmentation

For action trajectories, augmentations include:

- Temporal subsampling: Randomly sampling different control frequencies from the same trajectory;
- Noise injection: Adding small perturbations to action values;
- Trajectory interpolation: Interpolating between waypoints to generate smoother trajectories.

5.5 Fine-Tuning Strategies

5.5.1 Full Fine-Tuning

Full fine-tuning updates all parameters of the VLA model on robot data. This approach, used by RT-2 [Brohan et al., 2023a], achieves the best performance but requires significant computational resources (fine-tuning a 55B-parameter model requires multiple TPU/GPU pods).

5.5.2 Parameter-Efficient Fine-Tuning (PEFT)

To reduce the computational cost of fine-tuning, parameter-efficient methods have been adopted:

- **LoRA** [Hu et al., 2022a]: Adds low-rank adaptation matrices to selected weight matrices, updating only a small fraction of parameters. OpenVLA [Kim et al., 2024] uses LoRA to fine-tune its 7B model on consumer GPUs.
- **Adapter layers**: Inserts small trainable modules between transformer layers.
- **Prompt tuning**: Prepends learnable "soft prompts" to the input sequence that steer the model's behavior.

5.5.3 Multi-Stage Training

Several systems employ multi-stage training:

1. **Stage 1 (Pre-training):** Pre-train the VLM backbone on internet-scale vision-language data.
2. **Stage 2 (VLA fine-tuning):** Fine-tune on robot demonstration data to learn action prediction.

3. **Stage 3 (Task-specific adaptation):** Further fine-tune on specific tasks or embodiments using small amounts of data.

GR-2 [Bao et al., 2025] adds a video pre-training stage between stages 1 and 2, learning to predict future video frames before fine-tuning for action prediction.

5.6 Scaling Laws and Compute Considerations

5.6.1 Model Scaling

The VLA community is actively exploring how model scale affects performance. Key findings include:

- RT-2 [Brohan et al., 2023a] showed that a 55B VLA model significantly outperforms a 35M model (RT-1) on semantic generalization tasks, with the gap increasing for tasks requiring world knowledge.
- OpenVLA [Kim et al., 2024] demonstrated that a 7B model achieves competitive performance with proper training, suggesting that architectural innovations can compensate for scale.
- π_0 [Black et al., 2024] showed that flow matching enables effective action generation even at moderate model sizes (3B+), achieving performance that rivals or exceeds larger autoregressive models.

5.6.2 Data Scaling

Data scaling has proven at least as important as model scaling:

- RT-X [Collaboration et al., 2024] demonstrated consistent improvement as more diverse robot data was added to the training set.
- GR-2 [Bao et al., 2025] showed that pre-training on 38M video clips significantly improves manipulation performance.
- Physical Intelligence reported training π_0 on the largest known robot dataset, enabling unprecedented generalization.

5.6.3 Inference Efficiency

Real-world deployment of VLA models requires real-time inference (typically 5–20 Hz control rate). Key approaches to achieving this include:

- Action chunking: Executing a sequence of H actions from a single forward pass, reducing the required inference frequency.
- Quantization: Reducing model precision from float32 to int8 or int4.
- Distillation: Training a smaller "student" model to mimic the behavior of a larger "teacher."
- Caching: Reusing visual features across consecutive timesteps when the camera view changes slowly.

5.7 Summary

In this section, we have examined the training paradigms that underpin VLA models, covering objectives (autoregressive, diffusion, flow matching), datasets (robot demonstrations, vision-language data, video data), data collection strategies, augmentation techniques, fine-tuning approaches, and scaling considerations. The trend is toward larger models trained on more diverse data, with architectural innovations (flow matching, action chunking) that improve sample efficiency and inference speed. The next section turns to the complementary paradigm of world models.

6 World Models: Theory and Foundations

World models—neural networks that learn to predict the future dynamics of environments—represent a fundamental building block for intelligent agents. In this section, we establish the theoretical foundations of world models, tracing their evolution from classical dynamical systems theory through model-based reinforcement learning to modern generative world simulators.

6.1 Defining World Models

6.1.1 Formal Definition

A world model is a learned function $\hat{T}_\phi$ that approximates the true environment dynamics T. Given the current state (or observation) s_t and action a_t, the world model predicts the next state (or observation):

$$\hat{s}_{t+1} = \hat{T}_\phi(s_t, a_t) \tag{23}$$

In the stochastic case, the world model defines a distribution over next states:

$$\hat{s}_{t+1} \sim \hat{T}_\phi(\cdot \mid s_t, a_t) \tag{24}$$

More generally, world models can predict sequences of future states conditioned on action sequences:

$$\hat{s}_{t+1:t+H} \sim \hat{T}_\phi(\cdot \mid s_t, a_{t:t+H-1}) \tag{25}$$

The world model is typically trained to minimize a prediction objective:

$$\mathcal{L}_{WM}(\phi) = \mathbb{E}_{(s_t,a_t,s_{t+1})\sim\mathcal{D}}\left[d(s_{t+1}, \hat{T}_\phi(s_t, a_t))\right] \tag{26}$$

where $d(\cdot,\cdot)$ is a distance or divergence measure appropriate for the state space (e.g., MSE for continuous states, cross-entropy for discrete states, or perceptual loss for images).

6.1.2 Latent vs. Observation Space Prediction

A critical design choice is whether the world model operates in the raw observation space (e.g., pixels) or a learned latent space:

Observation-space world models predict future observations directly:

$$\hat{o}_{t+1} = \hat{T}_\phi(o_t, a_t) \tag{27}$$

These models, exemplified by UniSim [Yang et al., 2024] and Genie [Bruce et al., 2024], are directly interpretable and can be used for visual planning and simulation. However, predicting high-dimensional observations (e.g., 256×256 RGB images) is computationally expensive.

Latent-space world models first encode observations into a compact latent representation $z_t = E_\psi(o_t)$, then predict future latent states:

$$\hat{z}_{t+1} = \hat{T}_\phi(z_t, a_t) \tag{28}$$

These models, exemplified by the Dreamer family [Hafner et al., 2020, 2021, 2023], are more computationally efficient and can capture abstract dynamics that are easier to plan over. The trade-off is that the latent predictions may not be visually interpretable.

6.1.3 Stochastic vs. Deterministic Models

Real-world dynamics are inherently stochastic—the same action can lead to different outcomes depending on unobserved factors. World models must capture this stochasticity to be useful for planning. The two dominant approaches are:

1. **Variational models:** Introduce a latent variable z_t that captures stochastic transitions:

$$z_t \sim q_\phi(z_t \mid z_{t-1}, a_{t-1}, o_t), \quad \hat{o}_t = D_\phi(z_t) \tag{29}$$

   The posterior q_ϕ is trained to maximize the evidence lower bound (ELBO):

$$\mathcal{L}_{ELBO} = \mathbb{E}_{q_\phi}\left[\log p_\phi(o_t \mid z_t)\right] - \beta \cdot KL(q_\phi(z_t \mid z_{t-1}, a_{t-1}, o_t) \| p_\phi(z_t \mid z_{t-1}, a_{t-1})) \tag{30}$$

   where β controls the trade-off between reconstruction quality and latent space regularity. Dreamer [Hafner et al., 2020] and its successors use this approach.

2. **Diffusion-based models:** Model the distribution over future observations or latents using the diffusion framework:

$$o_{t+1}^{(k-1)} = \mu_\theta(o_{t+1}^{(k)}, k, o_t, a_t) + \sigma_k \epsilon \tag{31}$$

   This naturally handles multimodal predictions but requires iterative sampling. UniSim [Yang et al., 2024] and DIAMOND [Alonso et al., 2024] use diffusion-based world models.

6.2 Historical Evolution

6.2.1 Classical Model-Based RL

The concept of learning environment dynamics has deep roots in control theory and reinforcement learning. Sutton [1991] introduced the Dyna architecture, which integrates learning, planning, and reacting by learning a world model and using it to generate simulated experience for value function updates. The Dyna update rule combines real and simulated experience:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a)\right] \tag{32}$$

where $s' \sim \hat{T}(s, a)$ can come from either real interaction or model-predicted simulation.

Deisenroth et al. [2013] provided a comprehensive survey of model-based RL, categorizing approaches by how the learned model is used (for planning, data augmentation, or policy search) and the class of dynamics models (linear, Gaussian process, neural network).

6.2.2 Neural Network Dynamics Models

The application of neural networks to dynamics modeling gained traction with several influential works:

PILCO [Deisenroth and Rasmussen, 2011] used Gaussian processes for dynamics learning, providing uncertainty estimates that enable data-efficient policy search. While principled, Gaussian processes scale poorly with data size.

World Models (Ha and Schmidhuber) [Ha and Schmidhuber, 2018] demonstrated that a simple VAE-based world model trained in a compressed latent space can be used to train an agent entirely in "dreams." The architecture consists of three components: (1) a VAE that compresses observations into a latent representation; (2) an LSTM that predicts future latent states given actions; and (3) a simple linear controller that selects actions based on the latent state. This work popularized the concept of "dreaming" in world models.

PlaNet [Hafner et al., 2019] introduced the recurrent state-space model (RSSM), a stochastic recurrent model that combines deterministic and stochastic state transitions. The RSSM forms the basis of the Dreamer family:

$$h_t = f_\phi(h_{t-1}, s_{t-1}, a_{t-1}) \qquad \text{(deterministic transition)} \tag{33}$$

$$s_t \sim q_\phi(s_t \mid h_t, o_t) \qquad \text{(stochastic posterior)} \tag{34}$$

$$\hat{o}_t = g_\phi(s_t, h_t) \qquad \text{(observation reconstruction)} \tag{35}$$

where h_t is a deterministic recurrent state, s_t is a stochastic state, and q_ϕ is the posterior distribution.

6.2.3 The Dreamer Family

The Dreamer series represents the most influential line of work in latent-space world models for RL:

Dreamer (V1) [Hafner et al., 2020] learns a world model using the RSSM and trains a policy entirely within the learned latent space using actor-critic methods. The key innovation is that the agent never interacts with the real environment during policy training—all learning happens through "imagined" rollouts generated by the world model. Dreamer achieves state-of-the-art performance on 20 continuous control benchmarks with 2x data efficiency compared to prior methods.

DreamerV2 [Hafner et al., 2021] introduces several improvements: (1) categorical latent variables instead of Gaussian, which better capture discontinuous dynamics; (2) KL balancing, a modified ELBO objective that prevents the posterior from collapsing; and (3) a separate dynamics predictor that predicts future latents without observation reconstruction. DreamerV2 is the first world model to achieve human-level performance on the Atari 55 benchmark, demonstrating that latent world models can scale to visually complex environments.

DreamerV3 [Hafner et al., 2023] further improves with: (1) a symlog loss function that handles different magnitudes of prediction targets; (2) free bits for the KL regularizer to prevent information bottlenecks; and (3) a robust return normalization scheme. DreamerV3 achieves state-of-the-art across 150 diverse tasks (Atari, DMC, Minecraft, BabyAI) with fixed hyperparameters, demonstrating unprecedented generality for a model-based RL agent. Notably, DreamerV3 is the first algorithm to collect diamonds in Minecraft from scratch without human data.

6.3 Theoretical Properties of World Models

6.3.1 Approximation Theory

The universal approximation theorem guarantees that sufficiently large neural networks can approximate any continuous dynamics function $\hat{T} : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ to arbitrary precision. However, the sample complexity of learning accurate world models depends on:

- The *effective dimensionality* of the state space (which latent models reduce);
- The *stochasticity* of the dynamics (which variational models capture);
- The *temporal horizon* of predictions (which autoregressive models extend);
- The *partial observability* of the environment (which recurrent models address).

6.3.2 Model Errors and Compounding

A fundamental challenge for world models is error compounding: prediction errors accumulate over longer horizons, making long-range predictions unreliable. If the one-step prediction error is ϵ, the H-step error can grow as:

$$\|\hat{s}_{t+H} - s_{t+H}\| \leq \sum_{i=0}^{H-1} L^i \cdot \epsilon \tag{36}$$

where L is the Lipschitz constant of the dynamics. If $L > 1$ (chaotic dynamics), errors grow exponentially. Mitigation strategies include:

- Planning with short horizons and frequent replanning (MPC) [Chua et al., 2018];
- Ensemble models that quantify uncertainty [Chua et al., 2018, Janner et al., 2020];
- Latent space prediction, where the learned representation may have smoother dynamics than the observation space;
- Diffusion-based models that can capture multimodal predictions, reducing the impact of averaging errors.

6.3.3 Value-Aware Model Learning

Standard world model training minimizes prediction error on observed transitions, but this may not align with the goal of using the model for policy optimization. Value-aware model learning [Farahmand, 2017] proposes to weight the model's training objective toward transitions that are relevant for value estimation:

$$\mathcal{L}_{VA}(\phi) = \mathbb{E}\left[\|\hat{T}_\phi(s,a) - T(s,a)\|^2 \cdot \nabla_a Q(s,a)\right] \tag{37}$$

This focuses model capacity on transitions that matter for decision-making, potentially improving the model's utility for planning even when the overall prediction error is larger.

6.4 World Models as Simulators

A growing perspective views world models not just as tools for planning within RL, but as *generative simulators* of environments. This perspective, championed by systems like UniSim [Yang et al., 2024], Genie [Bruce et al., 2024], and Sora [OpenAI, 2024], treats world models as learned physics engines that can generate realistic environment dynamics for training, testing, and evaluation of embodied agents.

This "world model as simulator" perspective has several theoretical implications:

1. **Fidelity requirement:** The world model must generate sufficiently realistic dynamics for transfer to the real environment. This is a stronger requirement than merely supporting planning within RL.

2. **Controllability:** The world model must respond correctly to actions, enabling interactive simulation. This requires action-conditioned generation, which is more challenging than unconditional video prediction.

3. **Coverage:** The world model must cover the distribution of states and transitions that the agent will encounter, including rare or adversarial events.

4. **Consistency:** The world model must maintain physical consistency (e.g., object permanence, gravity, collision dynamics) across long generations.

6.5 Connections to VLA Models

World models and VLA models are theoretically complementary:

- **World models provide pre-training for VLA:** Learning to predict future observations given actions teaches the model about physical dynamics, which is useful for action prediction. GR-2 [Bao et al., 2025] and UniVLA [Li et al., 2025] exploit this connection.

- **VLA models provide action representations for world models:** The action space of a world model must be defined, and VLA-style action representations (continuous, discretized, or flow-matched) can be used.

- **Joint training:** A single model can be trained to both predict future states and generate actions, sharing representations and potentially improving both capabilities.

- **Planning with VLA:** A VLA model can be combined with a world model for model-predictive control: the world model generates candidate futures, and the VLA model evaluates or selects among them.

6.6 Summary

In this section, we have established the theoretical foundations of world models, covering formal definitions, historical evolution from Dyna to DreamerV3, theoretical properties (approximation, error compounding, value-aware learning), the emerging "world model as simulator" perspective, and connections to VLA models. The next section examines the technical approaches and systems that instantiate these theoretical ideas.

7 World Models: Techniques and Systems

In this section, we review the technical approaches and systems that instantiate the world model theory discussed in 6. We organize our review by the primary application domain: world models for reinforcement learning (enabling planning and imagination-augmented policy learning), world models for visual simulation (generating realistic environment rollouts), and world models for game environments (interactive simulation and game generation).

7.1 World Models for Reinforcement Learning

7.1.1 Model-Based Planning with Ensembles

Ensemble methods improve the robustness of world model-based planning by maintaining multiple models and using their disagreement as an uncertainty estimate. PETS [Chua et al., 2018] (Probabilistic Ensembles for Trajectory Sampling) combines an ensemble of probabilistic neural network dynamics models with model-predictive control (MPC). At each timestep, PETS:

1. Samples multiple trajectories using the ensemble of models;
2. Selects the action that maximizes the expected return under the trajectory distribution; and
3. Executes the first action and replans at the next timestep.

The ensemble captures both aleatoric uncertainty (inherent stochasticity) and epistemic uncertainty (model uncertainty due to limited data), enabling the agent to avoid actions with unpredictable outcomes.

MBPO [Janner et al., 2020] (Model-Based Policy Optimization with Dynamically Terminated Episodes) uses an ensemble of dynamics models to generate short synthetic rollouts (typically 1–5 steps) that augment real experience for training a model-free policy (SAC [Haarnoja et al., 2018]). The key insight is that short model rollouts are more reliable than long ones (due to error compounding), and even short rollouts provide significant data augmentation benefits. MBPO achieves 200x sample efficiency improvement over pure model-free methods on continuous control benchmarks.

TD-MPC [Hansen et al., 2022] and its successor TD-MPC2 [Hansen et al., 2024] combine model-based planning with model-free learning through a latent dynamics model. The world model predicts future latent states, which are used for both planning (via trajectory optimization) and value function learning. TD-MPC2 demonstrates that a single algorithm with fixed hyperparameters achieves strong performance across 80 continuous control tasks spanning locomotion, manipulation, and driving.

7.1.2 Latent Dynamics Models: The Dreamer Family

As discussed in 6, the Dreamer family represents the state of the art in latent-space world models for RL. We provide additional technical details here.

RSSM Architecture. The Recurrent State-Space Model (RSSM) [Hafner et al., 2019] combines deterministic and stochastic state transitions:

$$\text{Deterministic:} \quad h_t = \mathrm{GRU}(h_{t-1}, s_{t-1}, a_{t-1}) \tag{38}$$

$$\text{Stochastic (prior):} \quad \tilde{s}_t \sim p_\phi(\tilde{s}_t \mid h_t) \tag{39}$$

$$\text{Stochastic (posterior):} \quad s_t \sim q_\phi(s_t \mid h_t, \mathrm{Enc}(o_t)) \tag{40}$$

During training, the posterior uses the actual observation, while during imagination, only the prior is available. The gap between prior and posterior (measured by KL divergence) encourages the model to make accurate predictions.

DreamerV3's Symlog Predictions. A key innovation in DreamerV3 [Hafner et al., 2023] is the symlog loss function, which handles the challenge of predicting targets with vastly different magnitudes (e.g., small pixel changes vs. large reward values). The symlog function compresses large values while preserving small ones:

$$\text{symlog}(x) = \text{sign}(x) \ln(|x| + 1) \tag{41}$$

The model is trained to predict the symlog of the target, and predictions are converted back using the inverse function (symexp).

Scaling DreamerV3. DreamerV3 demonstrates that a single set of hyperparameters works across diverse domains:

- Atari 55 (discrete actions, visual observations): matches human-level performance;
- DeepMind Control (continuous actions, visual/low-dim observations): state-of-the-art data efficiency;
- Minecraft (sparse rewards, visual observations): first algorithm to obtain diamonds from scratch;
- BabyAI (language-conditioned navigation): strong performance on instruction-following tasks;
- DMLab (3D navigation with memory): competitive with specialized methods.

7.1.3 Transformer-Based World Models for RL

IRIS [Micheli et al., 2023] replaces the recurrent state-space model with a transformer-based world model. The architecture uses a VQ-VAE [van den Oord et al., 2017] to tokenize observations into discrete tokens and an autoregressive transformer (similar to GPT) to predict future tokens conditioned on actions. The policy is trained using the actor-critic algorithm within the imagined world.

IRIS achieves strong performance on Atari, with some advantages over DreamerV2 on games requiring longer temporal reasoning. However, the discrete tokenization can lose fine-grained visual information compared to continuous latent models.

STORM [Shi et al., 2024] extends the transformer-based approach with stochastic latent variables, combining the benefits of transformers (long-range temporal reasoning) with stochasticity (capturing uncertainty). STORM demonstrates improved performance on both Atari and DeepMind Control compared to both recurrent and deterministic transformer world models.

7.1.4 TD-MPC2: Scalable Model-Based RL

TD-MPC2 [Hansen et al., 2024] introduces a scalable model-based RL algorithm that combines:

1. A latent dynamics model that predicts future states in a learned representation space;
2. A model-predictive control planner that optimizes action sequences using the latent model;
3. A model-free value function that bootstraps the planner;

4. A task-inferred mechanism for multi-task learning.

TD-MPC2 is evaluated on 80 tasks spanning 4 domains, demonstrating that a single algorithm with fixed hyperparameters can achieve strong performance across diverse continuous control problems. The model-based component provides data efficiency, while the model-free component provides asymptotic performance.

7.2 World Models for Visual Simulation

7.2.1 UniSim: Learning Interactive Real-World Simulators

UniSim [Yang et al., 2024] represents a landmark in observation-space world models, learning a unified simulator of real-world visual interactions. Given an initial observation (image), a language instruction, and an action sequence, UniSim generates a video depicting the predicted outcome. The key architectural choices include:

1. A pre-trained T5 [Raffel et al., 2020] language encoder for processing instructions;
2. A ViT-based visual encoder for extracting image features;
3. A diffusion-based video generation model conditioned on actions and instructions;
4. Temporal attention layers for maintaining consistency across frames.

UniSim is trained on a diverse dataset of real-world interaction videos from driving, manipulation, and navigation scenarios. It demonstrates the ability to generate realistic videos of robot actions, driving maneuvers, and navigation episodes, enabling visual planning and evaluation of agent behaviors without physical execution.

7.2.2 Genie: Generative Interactive Environments

Genie [Bruce et al., 2024] learns to generate interactive 2D game environments from unlabeled internet videos. Given an initial frame and a sequence of actions ("latent actions"), Genie generates subsequent frames depicting the environment's response. The key innovation is that Genie learns the action space *unsupervised* from video data—it infers a discrete latent action space by training an inverse dynamics model on video pairs.

The architecture consists of:

1. A video tokenizer that encodes frames into discrete tokens;
2. A latent action model that infers actions between consecutive frames;
3. A dynamics model (autoregressive transformer) that predicts future tokens given current tokens and latent actions;
4. A video detokenizer that generates frames from tokens.

Genie can generate diverse, interactive 2D environments from a single prompt image, demonstrating zero-shot generation of playable games. While limited to 2D pixel art style, Genie represents a proof of concept for learning interactive simulators entirely from passive video data.

7.2.3 Genie 2: Foundation World Model

Genie 2 [DeepMind, 2024] extends Genie to 3D environments with photorealistic visual quality. Genie 2 generates interactive 3D environments from a single image prompt, responding to keyboard and mouse inputs in real-time. Key capabilities include:

- Generating diverse 3D environments (indoor, outdoor, urban, natural);
- Responding to user inputs for navigation and interaction;
- Maintaining object permanence and physical consistency over short horizons;
- Supporting diverse visual styles (photorealistic, stylized, cartoon).

Genie 2 is positioned as a "foundation world model"—a general-purpose environment generator that can be used for training embodied agents, testing behaviors, and creating interactive experiences. The model uses a large-scale diffusion-based video generation architecture with action conditioning.

7.2.4 NVIDIA Cosmos: World Foundation Models

NVIDIA Cosmos [NVIDIA, 2025] is a family of world foundation models designed for physical AI applications. Built on a diffusion transformer (DiT) architecture, Cosmos generates physically accurate videos conditioned on text prompts, images, and actions. Key features include:

- Multiple model sizes (from 4B to 50B+ parameters);
- Training on a curated dataset of physical world videos emphasizing physical accuracy;
- Support for camera control, object manipulation, and navigation actions;
- Integration with NVIDIA's Omniverse platform for synthetic data generation;
- Post-training alignment for physical consistency and controllability.

Cosmos is designed for applications in autonomous driving (generating driving scenarios), robotics (simulating manipulation and navigation), and digital twins (creating interactive virtual environments). It represents the industrial-scale realization of the world model paradigm, with a focus on practical deployment.

7.2.5 Sora as a World Model

OpenAI's Sora [OpenAI, 2024], while primarily presented as a video generation model, exhibits properties consistent with being a world model. Sora generates photorealistic videos up to 60 seconds long conditioned on text prompts, and the generated videos demonstrate an understanding of physical dynamics (object interactions, gravity, fluid dynamics). However, Sora also exhibits common world model failure modes: physical inconsistencies in complex scenes, violations of object permanence, and implausible dynamics in adversarial scenarios.

Sora's architecture is based on a diffusion transformer that operates on spatiotemporal patches of video, analogous to how ViT processes spatial patches of images. The model uses a video compression network to reduce dimensionality and a transformer decoder to generate compressed video tokens.

7.2.6 GAIA-1: World Models for Autonomous Driving

GAIA-1 [Hu et al., 2023a] is a world model specifically designed for autonomous driving. It generates driving videos conditioned on text prompts, actions (steering, acceleration), and structured inputs (maps, bounding boxes). The architecture consists of:

1. A video tokenizer that encodes driving scenes into discrete tokens;

2. An autoregressive transformer that predicts future tokens conditioned on actions and text;

3. A video decoder that generates driving scene video.

GAIA-1 demonstrates the ability to generate diverse driving scenarios (urban, highway, day, night, rain, clear) and respond realistically to driving actions (turning, braking, accelerating). It can be used for generating training data for perception systems, testing driving policies, and simulating rare safety-critical scenarios.

7.3 World Models for Game Environments

7.3.1 GameNGen: Diffusion-Based Game Engine

GameNGen [Valevski et al., 2024] demonstrates that a neural network can simulate a complete game environment in real-time. Trained on gameplay data from DOOM, GameNGen generates each frame conditioned on the previous frame and the player's action (keyboard/mouse input), producing playable game sessions at 20+ FPS.

The architecture uses a modified Stable Diffusion [Rombach et al., 2022] model with:

- Action conditioning through cross-attention layers;

- Temporal consistency via frame-to-frame conditioning;

- Augmented self-supervised training to reduce autoregressive drift.

GameNGen achieves a PSNR of 29.4 on generated gameplay frames, comparable to lossy JPEG compression. Human evaluators found it difficult to distinguish short GameNGen clips from real gameplay, demonstrating the visual quality of the simulation.

7.3.2 Oasis: Real-Time World Model

Oasis [AI and Etched, 2024] is a real-time interactive world model for Minecraft-like environments. Given a frame and player action, Oasis generates the next frame in real-time, creating an interactive game experience. The architecture uses a spatial autoencoder for efficient frame compression and a latent diffusion backbone for next-frame generation. Oasis runs at interactive frame rates (10+ FPS) on modern GPUs, making it one of the first practical neural game engines.

7.3.3 DIAMOND: Diffusion for RL in Atari

DIAMOND [Alonso et al., 2024] uses a diffusion-based world model for RL in Atari games. Rather than operating in a latent space, DIAMOND predicts next-frame pixel observations directly using a conditional diffusion model. The agent plans by generating imagined rollouts through the diffusion world model and selecting actions that maximize expected return.

Key technical innovations include:

- Using a smaller number of denoising steps (fewer than 10) for real-time planning;
- Conditioning on discrete actions through the diffusion process;
- Epsilon-greedy exploration within the imagined environment.

DIAMOND achieves competitive scores on a subset of Atari games, demonstrating that diffusion-based world models can support RL even in visually complex environments.

7.3.4 GenRL: World Models for RL Agent Training

GenRL [Flet-Berliac et al., 2024] trains world models on video data and uses them to train RL agents without access to the true environment. The approach involves:

1. Training a video generation model on gameplay videos;
2. Using the video model as a world model to generate training data for RL;
3. Transferring the RL agent from the video model to the real environment.

GenRL demonstrates that agents trained entirely in a video world model can achieve non-trivial performance when deployed in the real game, opening the possibility of training embodied agents from video data alone.

7.4 World Models with Memory and Long-Horizon Reasoning

7.4.1 R2I: Recurrent Reasoning and Imagination

R2I [Mo et al., 2024] introduces a recurrent reasoning module into world models, enabling multi-step imagination and planning. The model iteratively refines its prediction of future states by alternating between prediction and reasoning steps, achieving more accurate long-horizon predictions than single-pass models.

7.4.2 WM2: World Models with Memory

WM2 [Li et al., 2024c] incorporates an external memory module into the world model, enabling the agent to remember and reference past experiences during planning. The memory stores encoded state–action–outcome tuples and retrieves relevant experiences using attention. This is particularly valuable for tasks requiring long-term dependencies (e.g., "go to the room you visited earlier").

7.4.3 Motia: Multi-Task World Models

MOTIA [Gong et al., 2024] addresses the challenge of building multi-task world models that can simulate diverse environments with a single model. The key innovation is a task-conditioned architecture that adapts its dynamics predictions based on a task identifier or instruction, enabling a single world model to serve as a universal simulator across multiple tasks and environments.

7.5 LLM-Based World Models

An emerging paradigm leverages large language models as components of world models:

Table 4: Comparison of world model systems. "Latent" indicates latent-space prediction, "Pixel" indicates observation-space prediction, "AR" denotes autoregressive, and "Diff." denotes diffusion-based.

Model	Space	Gen.	Training Data	Application	Params	Key Innovation
DreamerV3 [Hafner et al., 2023]	Latent	VAE	Online RL	RL policy	400M	Fixed hyperparams
TD-MPC2 [Hansen et al., 2024]	Latent	Hybrid	Online RL	Multi-task RL	100M	Scalable planning
IRIS [Micheli et al., 2023]	Latent	AR	Online RL	RL (Atari)	200M	Transformer WM
STORM [Shi et al., 2024]	Latent	AR+VAE	Online RL	RL	300M	Stochastic transformer
UniSim [Yang et al., 2024]	Pixel	Diff.	Robot/driving video	Visual simulation	1B+	Real-world simulator
Genie [Bruce et al., 2024]	Pixel	AR	Internet video	Game generation	4B	Unsupervised actions
Genie 2 [DeepMind, 2024]	Pixel	Diff.	Internet video	3D world gen	Large	3D interactive
Cosmos [NVIDIA, 2025]	Pixel	Diff.	Physical videos	Physical AI	4B–50B+	Industrial scale
Sora [OpenAI, 2024]	Pixel	Diff.	Internet video	Video generation	Large	Long-form video
GameNGen [Valevski et al., 2024]	Pixel	Diff.	DOOM gameplay	Game engine	SD-based	Real-time game
Oasis [AI and Etched, 2024]	Pixel	Diff.	Minecraft	Game engine	Large	Interactive game
GAIA-1 [Hu et al., 2023a]	Pixel	AR	Driving video	Autonomous driving	Large	Driving simulator
DIAMOND [Alonso et al., 2024]	Pixel	Diff.	Online RL	RL (Atari)	200M	Diffusion RL

7.5.1 LLMs as Dynamics Predictors

Several works explore using LLMs to predict environment dynamics in text-based or structured representations. Notom et al. [2023] demonstrated that LLMs can simulate text-based game environments with high fidelity, generating coherent state descriptions conditioned on actions. Li et al. [2022] showed that LLMs can serve as reward functions for RL by evaluating the desirability of state transitions described in natural language.

7.5.2 LLM-Augmented Planning

LLMs can augment world model-based planning by:

- Decomposing high-level goals into subgoals that can be planned over [Huang et al., 2022];
- Generating candidate action sequences for the world model to evaluate [Hao et al., 2023];
- Providing commonsense constraints on physically plausible transitions [Ahn et al., 2022].

These approaches combine the semantic reasoning of LLMs with the physical simulation capabilities of world models.

7.6 Comparative Analysis of World Model Systems

4 provides a comprehensive comparison of world model systems.

7.7 Trends and Emerging Directions

Our analysis reveals several key trends in world model research:

1. **From latent to pixel space:** While early world models operated in learned latent spaces for efficiency, modern systems increasingly predict directly in pixel space, enabled by advances in diffusion models and computational scale. This trend is driven by the desire for interpretable, general-purpose simulators.

2. **From RL-specific to general-purpose:** World models are evolving from tools specifically designed for RL planning to general-purpose environment simulators that serve multiple purposes (training, evaluation, testing, entertainment).

3. **From single-task to multi-task/universal:** Modern world models aim to simulate diverse environments with a single model, moving toward the goal of a "universal world model."

4. **Convergence with video generation:** The architectures and training techniques of modern world models increasingly overlap with video generation models (diffusion transformers, temporal attention, text conditioning), suggesting convergence toward a unified paradigm.

5. **Scale as a key driver:** The performance of world models, particularly in pixel space, scales dramatically with model size and training data, following trends similar to those observed in language models.

7.8 Summary

In this section, we have reviewed the major technical approaches and systems for world models, covering latent-space models for RL (Dreamer family, TD-MPC2, IRIS, STORM), pixel-space models for visual simulation (UniSim, Genie, Cosmos, Sora), game-specific models (GameNGen, Oasis, DIAMOND), and emerging approaches incorporating memory and LLMs. The field is converging toward large-scale, general-purpose world models that can serve as universal simulators for embodied AI. The next sections examine how VLA models and world models are applied in real-world and virtual embodied AI settings.

8 Embodied AI in the Real World

In this section, we examine how VLA models and world models are deployed in real-world embodied AI applications. We cover robotic manipulation, mobile manipulation, dexterous manipulation, autonomous driving, navigation, and domain-specific applications including surgical robotics, agricultural robotics, and industrial automation. For each application domain, we discuss the unique challenges, key systems, and state-of-the-art performance.

8.1 Robotic Manipulation

Robotic manipulation—the ability to grasp, move, and interact with physical objects—is the primary application domain for VLA models. We organize our discussion by the complexity of the manipulation task.

8.1.1 Tabletop Manipulation

Tabletop manipulation involves a stationary robot arm operating on a table surface, performing tasks such as pick-and-place, stacking, pouring, and tool use. This is the most extensively studied setting for VLA models due to the relative simplicity of the environment and the availability of standardized benchmarks.

RT-1/RT-2 in Kitchen Environments. The RT-1 and RT-2 models [Brohan et al., 2023b,a] were evaluated on a suite of over 700 kitchen manipulation tasks, including picking up objects, opening drawers, placing objects in containers, and moving objects between locations. RT-2 demonstrated the ability to follow novel instructions involving objects never seen during robot training, such as "move the object near the extinct animal to the blue bowl," leveraging its VLM pre-training to identify a dinosaur toy as an "extinct animal."

OpenVLA on Bridge Data. OpenVLA [Kim et al., 2024] was primarily evaluated on the WidowX Bridge V2 benchmark [Walke et al., 2023], achieving state-of-the-art performance on

24 tabletop manipulation tasks. The model demonstrates strong generalization to new objects, backgrounds, and instruction phrasings, achieving 76.5% success rate on held-out tasks compared to 72.2% for RT-2-X (fine-tuned).

Diffusion Policy for Precise Manipulation. Diffusion Policy [Chi et al., 2023] excels at tasks requiring precise continuous control, such as pushing objects to specific locations, pouring, and multi-step assembly. Its ability to represent multimodal action distributions is particularly valuable for tasks where multiple strategies are valid (e.g., different grasps for an irregularly shaped object).

ACT and ALOHA for Bimanual Manipulation. The ACT policy [Zhao et al., 2023a], trained on the ALOHA platform, demonstrates impressive bimanual manipulation capabilities including:

- Sliding a zipper closed;
- Inserting a battery into a remote;
- Threading a cable through a clamp;
- Opening and closing a ziplock bag;
- Picking up and placing delicate objects (e.g., a grape).

These tasks require precise coordination between two arms, which is enabled by ACT's action chunking mechanism and the high-quality teleoperation data from ALOHA.

8.1.2 Bin Picking and Unstructured Manipulation

Bin picking—retrieving specific objects from a cluttered bin—is a fundamental industrial manipulation task with high practical value. VLA models applied to bin picking must handle severe occlusion, diverse object geometries, and clutter.

Chi et al. [2023] demonstrated that Diffusion Policy achieves near-perfect success rates on bin picking tasks with novel objects, outperforming traditional grasp planning pipelines. The key advantage is the ability to generate diverse grasping strategies that account for the specific arrangement of objects in the bin.

π_0 [Black et al., 2024] demonstrates bin picking as part of its general manipulation capabilities, including the ability to pick specific objects described in language ("pick up the red cube") from a bin containing multiple objects.

8.1.3 Long-Horizon Manipulation

Long-horizon manipulation tasks require executing sequences of primitive actions to achieve a high-level goal. Examples include:

- Setting a table (placing plates, utensils, glasses in correct positions);
- Assembling furniture (aligning parts, inserting screws);
- Cooking (chopping, mixing, transferring ingredients).

CogAct [Li et al., 2024b] addresses long-horizon manipulation through hierarchical decomposition: a VLM-based planner generates subgoals, and a diffusion-based executor achieves each subgoal. This approach achieves state-of-the-art performance on the CALVIN long-horizon benchmark [Mees et al., 2022], successfully completing sequences of up to 5 chained tasks.

GR-2 [Bao et al., 2025] demonstrates long-horizon manipulation through its video prediction pre-training, which enables the model to anticipate the outcomes of multi-step action sequences. The model successfully completes tasks like “put the block in the box and close the lid” that require temporal reasoning.

SayPlan [Rana et al., 2023] combines LLM-based planning with a world model for long-horizon task planning in large-scale environments (e.g., entire buildings). The LLM generates a high-level plan, and the world model simulates each step to verify feasibility before execution.

8.2 Mobile Manipulation

Mobile manipulation combines locomotion with manipulation, enabling robots to perform tasks that require moving through an environment and interacting with objects at different locations.

8.2.1 Mobile ALOHA

Mobile ALOHA [Fu et al., 2024] extends the ALOHA platform with a mobile base, enabling bi-manual manipulation while navigating through a kitchen environment. The system demonstrates tasks such as:

- Cooking and serving a stir-fry dish (requiring navigation to the stove, using a wok, and serving);
- Loading and unloading a dishwasher;
- Pushing a cart;
- Operating an elevator.

The key challenge is coordinating whole-body control (base + two arms + grippers) with long-horizon task planning. Mobile ALOHA uses imitation learning on whole-body demonstrations, achieving 80–90% success rates on complex multi-step tasks.

8.2.2 π_0 for Mobile Manipulation

π_0 [Black et al., 2024] demonstrates mobile manipulation capabilities including busing tables (navigating to a table, picking up dishes, and placing them in a bin), folding laundry, and assembling boxes. The flow-matching action representation handles the high-dimensional continuous action space of mobile manipulators effectively.

8.3 Dexterous Manipulation

Dexterous manipulation involves controlling multi-fingered hands to perform tasks requiring fine motor control, such as in-hand rotation, tool use, and object manipulation.

8.3.1 Shadow Hand Manipulation

Several works apply VLA-like approaches to dexterous manipulation with the Shadow Hand:

- **DexMV** [Qin et al., 2022]: Uses human hand motion as demonstrations, retargeted to the Shadow Hand through optimization-based mapping.

- **Dexterous Hand Manipulation with Diffusion Policy** [Chi et al., 2023]: Applies Diffusion Policy to in-hand rotation and reorientation tasks.
- **Functional Grasping** [Shaw et al., 2023]: Generates functionally appropriate grasps for tool use (e.g., grasping a hammer by the handle, not the head).

8.3.2 Sim-to-Real for Dexterous Manipulation

Sim-to-real transfer is particularly important for dexterous manipulation due to the difficulty of collecting real-world demonstrations. Key approaches include:

- Domain randomization in simulation [Tobin et al., 2017];
- Teacher-student distillation from privileged to sensor-based policies [Chen et al., 2020];
- Adaptive sim-to-real via system identification [Yu et al., 2017].

8.4 Autonomous Driving

Autonomous driving represents a major application domain for world models, with significant industry investment.

8.4.1 World Models for Driving Simulation

GAIA-1 [Hu et al., 2023a] generates realistic driving videos conditioned on actions and text prompts, enabling the simulation of diverse driving scenarios. The model can generate rare events (e.g., pedestrian crossing, sudden braking) that are critical for safety testing but difficult to encounter in real-world data collection.

MILE [Hu et al., 2022b] learns a world model from driving data that predicts future bird's-eye-view (BEV) representations, which are used for planning. The model combines a latent dynamics model with a decoder that generates BEV predictions, enabling model-based planning in a structured representation.

UniSim for Driving. UniSim [Yang et al., 2024] is applied to autonomous driving scenarios, generating realistic videos of driving maneuvers conditioned on ego-vehicle actions. This enables closed-loop simulation of driving policies, where the world model generates the environment's response to the driving policy's actions.

Cosmos for Autonomous Driving. NVIDIA Cosmos [NVIDIA, 2025] includes driving-specific world models that generate diverse driving scenarios for training and testing autonomous driving systems.

8.4.2 VLA Models for Driving

Several works apply VLA-style approaches to autonomous driving:

- **GPT-Driver** [Wei et al., 2023]: Uses GPT-4 for driving planning, treating driving as a language reasoning problem.
- **DriveGPT4** [Zheng et al., 2023]: Fine-tunes a VLM for interpreting driving scenes and generating driving actions.
- **UniAD** [Hu et al., 2023b]: A unified autonomous driving framework that performs perception, prediction, and planning in an end-to-end manner.

- **VAD** [Jiang et al., 2024]: Vectorized end-to-end autonomous driving using transformer-based planning.

8.5 Embodied Navigation

Navigation—the ability to move through an environment to reach a goal location—is another key application of embodied AI.

8.5.1 Visual Navigation

VLA models for navigation process visual observations and language instructions to generate navigation actions:

- **LM-Nav** [Shah et al., 2023a]: Uses large models for language-guided visual navigation in outdoor environments.
- **CoW** [Gadre et al., 2023]: Combines CLIP with a world model for zero-shot object-goal navigation.
- **NavGPT** [Zhou et al., 2023]: Uses GPT-4 for navigation planning with explicit reasoning.

8.5.2 World Models for Navigation

World models support navigation through:

- **Topological planning:** Building a graph of the environment from imagined rollouts [Shah et al., 2023b];
- **Terrain prediction:** Predicting future observations to evaluate candidate paths [Shah et al., 2023b];
- **Exploration:** Using uncertainty in the world model's predictions to guide exploration [Chaplot et al., 2020].

8.5.3 Indoor Navigation

Indoor navigation in homes and offices requires understanding room layouts, object locations, and semantic concepts:

- **HomeRobot** [Collaboration, 2023a]: An open-source platform for indoor navigation and interaction, combining mobile navigation with manipulation.
- **OK-Robot** [Liu et al., 2024b]: Combines open-vocabulary detection with navigation for zero-shot object retrieval in novel homes.
- **Robotouille** [Zhu et al., 2024]: A benchmark for long-horizon indoor manipulation and navigation.

8.6 Domain-Specific Applications

8.6.1 Surgical Robotics

Surgical robotics requires extreme precision and safety, making it a challenging domain for VLA models. Key works include:

- **SurgVLP** [Yuan et al., 2023]: Visual-language pre-training for surgical scene understanding.
- **Surgical VLA** [Li et al., 2024d]: Adapts VLA models for surgical instrument control.
- **World models for surgical simulation**: Generating realistic surgical scenarios for training surgical robots [Peters et al., 2023].

8.6.2 Agricultural Robotics

Agricultural applications include harvesting, weeding, and crop monitoring:

- **Robotic harvesting** [Zhao et al., 2023b]: Using VLA models for fruit picking with language-specified target fruits.
- **Weeding robots** [Lottes et al., 2020]: Navigation and manipulation in unstructured agricultural environments.
- **Crop monitoring** [Tseng et al., 2023]: Using world models for predicting crop growth and planning harvests.

8.6.3 Industrial Automation

In industrial settings, VLA models and world models are applied to:

- **Assembly**: Automating multi-step assembly tasks [Tian et al., 2024];
- **Quality inspection**: Using VLMs for visual inspection with natural language defect descriptions [Li et al., 2023];
- **Warehouse automation**: Picking, packing, and sorting in logistics facilities [Zeng et al., 2020];
- **Digital twins**: Using world models as digital twins of industrial processes for optimization and testing [NVIDIA, 2025].

8.7 Sim-to-Real Transfer in Practice

A critical challenge for all real-world applications is the sim-to-real gap. Modern approaches combine:

1. **World model simulation:** Training policies in world model-generated environments that approximate real-world dynamics [Yang et al., 2024];
2. **Domain randomization:** Varying visual and physical parameters during training to improve robustness [Tobin et al., 2017];
3. **Real-world fine-tuning:** Adapting simulation-trained policies with a small amount of real-world data [Chi et al., 2023];

4. **System identification:** Calibrating the simulation to match the real robot's dynamics [Yu et al., 2017].

The integration of VLA models (which can be fine-tuned with small amounts of real data) with world models (which can generate realistic training environments) represents the most promising path toward practical sim-to-real transfer.

8.8 Summary

In this section, we have surveyed the application of VLA models and world models across diverse real-world embodied AI domains, including robotic manipulation (tabletop, bin picking, long-horizon, mobile, dexterous), autonomous driving, navigation (visual, indoor), and domain-specific applications (surgical, agricultural, industrial). The key insight is that VLA models are moving beyond controlled laboratory settings toward practical deployment in unstructured, real-world environments, with world models playing an increasingly important role in training and evaluation. The next section examines applications in virtual worlds and games.

9 Embodied AI in Virtual Worlds and Games

Virtual worlds and games provide rich, controlled, and scalable environments for developing and testing embodied AI agents. In this section, we examine how VLA models and world models are applied in game AI agents, interactive game generation, game-based training environments, and the broader intersection of generative AI and interactive entertainment. These virtual environments serve both as end applications (creating intelligent game agents and game content) and as scalable testbeds for embodied AI research.

9.1 Game AI Agents

Game AI agents are embodied agents that operate within game environments, perceiving game states through visual or structured observations and generating actions (keyboard/mouse inputs or API commands) to achieve goals. The application of VLA and world model techniques to game AI has produced agents with increasingly sophisticated reasoning, planning, and interaction capabilities.

9.1.1 LLM-Based Game Agents

Large language models have emerged as powerful "brains" for game agents, providing natural language understanding, reasoning, and planning capabilities. We review the most influential LLM-based game agent systems.

VOYAGER [Wang et al., 2023a] is an LLM-powered agent for Minecraft that autonomously explores, learns skills, and accomplishes tasks. VOYAGER consists of three key components:

1. **Automatic curriculum:** An LLM generates exploration tasks based on the agent's current state and discovery progress, ensuring a steady progression of challenges.
2. **Skill library:** The agent maintains a growing library of reusable JavaScript skills (e.g., "craft stone sword," "build shelter") that are verified and stored for future use.
3. **Iterative prompting:** When a skill fails, the LLM receives an error message and generates a corrected version, enabling self-improvement.

VOYAGER achieves state-of-the-art performance on Minecraft technology tree milestones, obtaining unique items 15.3x faster than prior methods and unlocking the full technology tree in 130 iterations.

SPRING [Wang et al., 2023b] studies LLM agents playing the game "Crafter" by reading the game's Wikipedia page. The agent uses the game manual as a source of knowledge, extracting rules, strategies, and tips through a structured prompting approach. SPRING demonstrates that LLMs can acquire game-playing skills purely from reading about the game, achieving competitive performance without any gameplay experience.

STEVE-1 [Zhao et al., 2024] combines a pre-trained vision-language model with a reinforcement learning policy to create a Minecraft agent that follows natural language instructions. The system uses a VLM to interpret visual observations and language instructions, generating subgoals for a pre-trained RL policy. STEVE-1 demonstrates instruction-following in Minecraft across diverse tasks specified in natural language.

GitM (Ghost in the Minecraft) [Zhu et al., 2023] uses GPT-4 as a planner for Minecraft, decomposing high-level goals into executable action sequences through structured prompting. The system maintains a knowledge base of Minecraft recipes, crafting rules, and spatial information, enabling the LLM to reason about complex crafting chains.

JARVIS-1 [Wang et al., 2024c] extends the LLM agent paradigm with multimodal perception and long-term memory. The system uses a vision-language model for scene understanding, an LLM for planning, and a policy network for execution. A memory module stores past experiences (successful plans, failed attempts, explored areas) that are retrieved during planning.

9.1.2 VLM-Based Game Agents

Vision-language models enable game agents that can perceive game visuals directly (rather than relying on structured state information) and reason about game states in natural language.

ROCKET-1 [Tan et al., 2024a] introduces a novel approach where a VLA model is trained to play Minecraft from visual observations. The agent processes game screenshots and language instructions to generate mouse and keyboard actions, demonstrating that VLA architectures designed for robotics can be adapted for game playing.

Cradle [Tan et al., 2024b] is a general computer control agent that can play games through visual observation and keyboard/mouse input. The system uses a VLM for screen understanding and action generation, combined with a memory module for retaining task knowledge. Cradle has been demonstrated on games including Red Dead Redemption 2, Stardew Valley, and Drug Dealer Simulator.

RAGENT [Wang et al., 2024d] combines retrieval-augmented generation with LLM-based planning for game agents, maintaining a knowledge base of game-specific information that is dynamically retrieved during planning.

9.1.3 World Model-Based Game Agents

World models enable game agents to plan by simulating future game states mentally, without executing actions in the actual game.

DreamerV3 in Minecraft. DreamerV3 [Hafner et al., 2023] achieves the milestone of obtaining diamonds in Minecraft from scratch, entirely through world model-based imagination. The agent learns a latent world model from visual observations, imagines future states to train its policy, and never uses human demonstrations. This demonstrates that world models can learn complex, long-horizon game strategies from scratch.

IRIS in Atari. IRIS [Micheli et al., 2023] uses a transformer-based world model to achieve strong performance on Atari games. The agent learns to simulate game dynamics in imagination and trains its policy entirely within the imagined world, demonstrating the scalability of world model-based approaches to visually complex games.

DIAMOND in Atari. DIAMOND [Alonso et al., 2024] uses a diffusion-based world model for Atari, generating next-frame predictions conditioned on actions. The agent plans by generating imagined rollouts and selecting actions that maximize the expected return under the diffusion-predicted future.

Model-Based Planning in Strategy Games. Several works apply world models to strategy and board games:

- **MuZero** [Schrittwieser et al., 2020]: Combines a world model with Monte Carlo Tree Search (MCTS) for planning in board games (Go, Chess, Shogi) and Atari. MuZero learns the game dynamics and value function jointly, achieving superhuman performance without knowing the game rules.
- **Studento** [?]: Extends MuZero with policy distillation for efficient planning.
- **EZ-Greedy** [Danihelka et al., 2022]: Uses a world model for policy improvement in visual game environments.

9.2 Interactive Game Generation

A transformative application of world models is the generation of interactive game environments—neural networks that simulate entire games in real-time, responding to player inputs.

9.2.1 Diffusion-Based Game Engines

GameNGen [Valevski et al., 2024] represents a breakthrough in neural game simulation. By training a diffusion model on gameplay data from DOOM, the system generates each frame conditioned on the previous frame and the player's keyboard/mouse input. The result is a playable game that runs entirely through a neural network, achieving:

- Real-time generation at 20+ FPS;
- Visual quality comparable to the original game (PSNR 29.4);
- Correct game dynamics (enemy behavior, item pickup, door opening);
- Short-term consistency (maintaining scene coherence across 3–5 second clips).

The training process involves two phases: (1) a reinforcement learning agent is trained to play the game, collecting gameplay data; (2) a diffusion model is trained to predict the next frame given the current frame and action. An augmented loss function reduces autoregressive error accumulation by occasionally feeding the model's own predictions as input during training.

Oasis [AI and Etched, 2024] extends neural game simulation to Minecraft, generating interactive 3D voxel worlds from a single image prompt. The system runs at interactive frame rates and supports keyboard/mouse input for navigation and block placement. Oasis uses a spatial autoencoder for efficient frame compression and a latent diffusion backbone for frame prediction.

9.2.2 Procedural World Generation with World Models

World models can also be used for procedural generation of game content:

- **Genie** [Bruce et al., 2024]: Generates diverse 2D game environments from a single image, learning the dynamics unsupervised from internet videos. Each generated environment responds to player inputs, creating a unique interactive experience.
- **Genie 2** [DeepMind, 2024]: Extends to 3D environments, generating photorealistic interactive scenes from a single image prompt. The system supports navigation, object interaction, and diverse visual styles.
- **Custom world generation**: Using text-to-3D and video generation models for creating custom game levels and environments [Poole et al., 2023, Lin et al., 2023].

9.3 Game-Based Training Environments for Embodied AI

Games provide scalable, safe, and reproducible environments for training embodied AI agents. We review the major game-based training environments and their role in VLA and world model research.

9.3.1 Simulation Platforms

Minecraft is the most widely used game environment for embodied AI research, offering:

- Rich 3D voxel world with diverse biomes, structures, and resources;
- Open-ended gameplay with crafting, building, and survival mechanics;
- The MineDojo/MineRL [Fan et al., 2022] API for programmatic access;
- A large community and extensive documentation (Wikipedia, wikis).

Minecraft has been used to develop and evaluate world models (DreamerV3), LLM agents (VOYAGER, GitM, STEVE-1), and VLA-style agents (ROCKET-1).

Atari Learning Environment (ALE) [Bellemare et al., 2013] provides a standardized interface to Atari 2600 games, widely used for evaluating world models (DreamerV2/V3, IRIS, DIAMOND) and reinforcement learning algorithms.

Crafter [Hafner, 2022] is a procedurally generated 2D survival game inspired by Minecraft, designed as a benchmark for world model and RL research. It provides a standardized evaluation with 22 achievements across survival and technology tree progression.

NetHack Learning Environment (NLE) [Kuttler et al., 2020] provides access to the roguelike game NetHack, challenging agents with complex procedurally generated dungeons, long-horizon planning, and partial observability.

RoboCraft [Mu et al., 2023] is a robotic simulation platform built on game engine technology, providing realistic physics for robot manipulation research.

9.3.2 Navigation and Interaction Simulators

Habitat [Savva et al., 2019] is a simulation platform for embodied navigation research, providing photorealistic 3D environments (Gibson, Matterport3D, HM3D) and support for both point-goal and object-goal navigation.

AI2-THOR [Kolve et al., 2017] provides interactive 3D environments for embodied AI, supporting manipulation, navigation, and visual question answering in simulated households.

VirtualHome [Puig et al., 2018] simulates household activities, enabling agents to execute complex household tasks through structured action sequences.

RoboCasa [Bharadhwaj et al., 2024] is a large-scale simulation framework for robot learning in household environments, providing diverse scenes, objects, and tasks for training manipulation and navigation policies.

9.3.3 Sim-to-Real Transfer via Game Environments

Game environments play a critical role in sim-to-real transfer for embodied AI:

- **Isaac Gym** [Makoviychuk et al., 2021]: GPU-accelerated physics simulation for training robot policies that transfer to real hardware.
- **MuJoCo** [Todorov et al., 2012]: A physics engine widely used for robot simulation, providing accurate contact dynamics.
- **SAPIEN** [Xiang et al., 2020]: A simulation platform for part-level manipulation with realistic physics.
- **ManiSkill** [Gu et al., 2023]: A benchmark for robot manipulation in simulation with standardized evaluation.

World models enhance sim-to-real transfer by generating more realistic simulated experiences. For example, UniSim [Yang et al., 2024] can generate photorealistic simulated manipulation scenarios from real-world images, bridging the visual gap between simulation and reality.

9.4 LLM Agents as Game Designers and Storytellers

Beyond playing games, LLMs and world models are being used to *create* game experiences:

9.4.1 Procedural Narrative Generation

LLMs can generate branching narratives, dialogue, and quest structures for role-playing games:

- **DramaKeeper** [Liu et al., 2024c]: Uses LLMs to generate and maintain narrative consistency in interactive stories.
- **LLM as Dungeon Master** [Zhu, 2023]: Uses LLMs to run tabletop RPG sessions, generating narrative descriptions and managing game state.

9.4.2 World Building with Generative Models

Generative models enable rapid creation of game worlds:

- Text-to-3D models for generating game assets [Poole et al., 2023];
- Text-to-music for generating game soundtracks [Agostinelli et al., 2023];
- Diffusion models for generating game textures and sprites [Rombach et al., 2022].

Table 5: Comparison of game AI agent approaches across different game environments.

Agent	Game	Core Technique	Perception	Key Capability
VOYAGER [Wang et al., 2023a]	Minecraft	LLM + skill library	Text API	Autonomous exploration
DreamerV3 [Hafner et al., 2023]	Minecraft	Latent world model	Visual	Diamond from scratch
STEVE-1 [Zhao et al., 2024]	Minecraft	VLM + RL	Visual	Instruction following
MuZero [Schrittwieser et al., 2020]	Board games	World model + MCTS	Structured	Superhuman play
IRIS [Micheli et al., 2023]	Atari	Transformer WM	Visual	Imagination-based RL
DIAMOND [Alonso et al., 2024]	Atari	Diffusion WM	Visual	Diffusion planning
Cradle [Tan et al., 2024b]	Multiple	VLM	Visual	General computer control
SPRING [Wang et al., 2023b]	Crafter	LLM + manual	Visual	Learning from manuals
CICERO [Team et al., 2022]	Diplomacy	LLM + RL	Structured	Strategic communication

9.5 Open Multiplayer Environments

Multiplayer game environments present unique challenges for embodied AI, requiring social reasoning, collaboration, and adversarial interaction.

9.5.1 Social Game Agents

Generative Agents [Park et al., 2023] create believable simulacra of human behavior in a simulated town. Each agent is powered by an LLM that maintains a memory stream, reflects on past experiences, and plans daily activities. Agents interact with each other and the environment, producing emergent social behaviors such as organizing a Valentine's Day party.

Human-AI Collaboration in Games. Several works explore human-AI collaboration in game settings:

- **Hidden Society** [Gandhi et al., 2023]: Studies social reasoning in hidden-role games.
- **Overcooked-AI** [Carroll et al., 2019]: A benchmark for human-AI coordination in a cooperative cooking game.
- **CICERO** [Team et al., 2022]: An AI agent that achieves human-level performance in the strategic board game Diplomacy, combining language understanding with strategic planning.

9.6 Comparative Analysis of Game AI Approaches

5 provides a comparison of game AI agent approaches.

9.7 Key Challenges for Virtual World AI

9.7.1 Long-Term Consistency

Neural game engines and world models struggle with maintaining consistency over extended gameplay sessions. Objects may spontaneously appear or disappear, and the environment may forget past events. Approaches to addressing this include:

- External memory modules that store and retrieve game state [Li et al., 2024c];
- Hierarchical generation with a persistent high-level state and transient low-level details;
- Consistency losses that penalize deviations from the initial state.

9.7.2 Controllability

Ensuring that generated game content responds correctly to player actions is challenging. The world model must learn the causal relationship between actions and state changes, not just the statistical correlation. This is particularly difficult for:

- Rare actions that were underrepresented in training data;
- Long action sequences where errors compound;
- Actions that have delayed or indirect effects.

9.7.3 Scalability

Real-time interactive generation requires generating each frame within a tight time budget (typically 30–60 FPS for smooth gameplay). Current neural game engines achieve 10–20 FPS, limiting their application to games with lower frame rate requirements. Approaches to improving scalability include:

- Latent space generation with lightweight decoders;
- Model distillation and quantization;
- Temporal super-resolution (generating at low frame rate and interpolating).

9.8 Summary

In this section, we have examined the application of VLA models and world models in virtual worlds and games, covering game AI agents (LLM-based, VLM-based, world model-based), interactive game generation (GameNGen, Oasis), game-based training environments (Minecraft, Atari, Habitat), and emerging applications in procedural content generation and social simulation. Virtual environments serve as both testbeds for embodied AI research and end applications, with the boundary between AI research and game development becoming increasingly blurred.

10 Benchmarks, Datasets, and Evaluation

The evaluation of VLA models and world models requires carefully designed benchmarks, diverse datasets, and standardized metrics. In this section, we catalog the major benchmarks, discuss evaluation methodologies, and analyze the limitations of current evaluation practices.

10.1 Benchmarks for VLA Models

10.1.1 Real-World Manipulation Benchmarks

WidowX Bridge V2 [Walke et al., 2023] is the de facto standard benchmark for evaluating VLA models on real-world tabletop manipulation. It provides:

- 24 task categories with language instructions;
- Train/test splits with held-out objects and instructions;
- Evaluation protocol requiring 20 trials per task;

- Standardized hardware (WidowX 250 robot arm with wrist-mounted camera).

Key evaluation metrics include task success rate, generalization to novel objects, and generalization to novel instruction phrasings. OpenVLA [Kim et al., 2024] achieved 76.5% average success rate on the Bridge V2 held-out set.

SIMPLER [Li et al., 2024e] is a suite of reproducible real-world manipulation benchmarks designed for evaluating VLA models. It provides standardized task definitions, evaluation protocols, and data collection procedures across multiple robot platforms. SIMPLER includes tasks ranging from simple pick-and-place to complex multi-step manipulation, with both seen and unseen object configurations.

LIBERO [Liu et al., 2023b] is a benchmark for lifelong learning in robotic manipulation, consisting of 130 task descriptions organized into 10 task suites of increasing complexity. Tasks are performed on a Franka Panda robot in simulation and require the agent to learn and transfer across tasks.

ALOHA Tasks [Zhao et al., 2023a] include a set of bimanual manipulation tasks on the ALOHA platform:

- Simple tasks: Inserting a marker, opening a cup;
- Medium tasks: Sliding a zipper, threading a cable;
- Hard tasks: Preparing a serving, wrapping a chocolate.

Evaluation requires 20–25 trials per task with success rate as the primary metric.

SRLBench [Collaboration, 2023b] provides standard benchmarks for sim-to-real transfer evaluation, measuring the gap between simulation and real-world performance.

10.1.2 Simulated Manipulation Benchmarks

RLBench [James et al., 2020] provides 100 manipulation tasks in a simulated environment with diverse objects, goal conditions, and language descriptions. Each task includes 100 scripted demonstrations for behavioral cloning. RLBench is widely used for evaluating multi-task manipulation policies.

CALVIN [Mees et al., 2022] is a benchmark for long-horizon language-conditioned manipulation in a simulated tabletop environment. It provides:

- 34 manipulation tasks organized into 4 difficulty levels;
- Evaluation on composing sequences of up to 5 tasks (Longest Instruction Chain);
- Both static and play data for training;
- Standardized evaluation protocol across 4 environment splits.

CALVIN is the primary benchmark for evaluating long-horizon VLA models. GR-1 [Huang et al., 2024] and CogAct [Li et al., 2024b] achieve state-of-the-art results on CALVIN.

ManiSkill2 [Gu et al., 2023] provides 200+ manipulation tasks across 20 task categories, supporting both rigid and soft-body manipulation. It includes standardized baselines and evaluation protocols.

RoboGen [Wang et al., 2024e] generates diverse simulated manipulation tasks procedurally using LLMs, providing unlimited task variety for training and evaluation.

10.1.3 Mobile Manipulation Benchmarks

HomeRobot [Collaboration, 2023a] evaluates mobile manipulation in home environments, requiring navigation to objects, picking them up, and placing them in target locations.

Stretch RE1 Tasks [Robot, 2020] provide benchmarks on the Hello Robot Stretch RE1, a low-cost mobile manipulator for home environments.

10.2 Benchmarks for World Models

10.2.1 RL Benchmarks for World Models

Atari 55/100 [Bellemare et al., 2013, Hafner et al., 2021]: A set of Atari 2600 games used to evaluate world model-based RL agents. Performance is measured as the percentage of human-normalized score. DreamerV2 was the first world model to achieve human-level performance on the Atari 55 benchmark.

DeepMind Control Suite [Tassa et al., 2018]: A set of continuous control tasks with visual observations, widely used for evaluating world model-based RL. Performance is measured as cumulative reward over training.

BSuite [Osband et al., 2020]: A benchmark suite for evaluating RL agents' basic capabilities (generalization, exploration, credit assignment), useful for analyzing world model behavior.

Rlax Benchmarks: Google DeepMind's RL benchmarks for evaluating model-based methods.

10.2.2 Video Prediction Benchmarks

Kinetics [Kay et al., 2017]: A large-scale video dataset for evaluating video prediction quality, measuring FID, FVD, and IS metrics.

UCF-101 [Soomro et al., 2012]: A video action recognition dataset repurposed for evaluating video generation quality.

BAIR Robot Pushing [Finn et al., 2016]: A video prediction benchmark featuring a robot pushing objects on a table, with standardized evaluation metrics (PSNR, SSIM, FVD).

10.2.3 Game Simulation Benchmarks

DOOM (GameNGen) [Valevski et al., 2024]: Evaluated on gameplay footage from DOOM, measuring PSNR, SSIM, and playable FPS.

Minecraft (Oasis) [AI and Etched, 2024]: Evaluated on Minecraft gameplay, measuring visual quality and controllability.

Genie Benchmark [Bruce et al., 2024]: A benchmark for evaluating generative interactive environments, measuring generation quality, controllability, and diversity.

10.3 Evaluation Metrics

10.3.1 Task Performance Metrics

- **Success Rate (SR):** The fraction of task attempts that achieve the goal state. The most common metric for VLA evaluation.
- **Average Success Rate (ASR):** Mean success rate across multiple task categories.
- **Generalization metrics:** Performance on held-out objects, instructions, or environments that were not seen during training.

Table 6: Comprehensive list of major robot demonstration datasets.

Dataset	**Robot(s)**	**Episodes**	**Tasks**	**Year**	**Modality**	**Source**
RT-1 Dataset	Kuka, etc.	130k	700+	2023	Image, lang, action	Google
Open X-Embodiment	22 robots	1M+	500+	2024	Image, lang, action	13 inst.
Bridge V2	WidowX	24k	24	2023	Image, lang, action	UC Berkeley
DROID	Franka	76k	56	2024	Multi-view, action	Multiple
CALVIN	Franka (sim)	24h play	34	2022	Image, lang, action	NVIDIA
ALOHA	ALOHA bimanual	Varies	Varies	2023	Image, action	Stanford
Ego4D	Human hands	3.67M clips	–	2022	Ego video, narr.	Meta
Something-Something	Human hands	220k	174	2017	Video, labels	20BN

- **Instruction following accuracy:** Whether the agent correctly follows the specified instruction (vs. completing the task through an alternative strategy).

10.3.2 World Model Quality Metrics

- **Fréchet Video Distance (FVD):** The standard metric for video generation quality, measuring the distance between the distributions of real and generated videos in a feature space.
- **Peak Signal-to-Noise Ratio (PSNR):** Measures pixel-level reconstruction quality.
- **Structural Similarity Index (SSIM):** Measures perceptual image quality.
- **Inception Score (IS):** Measures the diversity and quality of generated samples.
- **Fréchet Inception Distance (FID):** Measures the quality of individual generated frames.
- **Action-conditioned metrics:** For world models, additional metrics measure how well the generated content responds to actions (e.g., action prediction accuracy, trajectory likelihood).

10.3.3 Agent-Level Metrics

- **Sample efficiency:** How many environment interactions are needed to achieve a target performance level.
- **Data efficiency:** How many demonstrations are needed for VLA fine-tuning.
- **Inference latency:** Time from observation to action prediction, critical for real-time control.
- **Zero-shot transfer:** Performance on tasks, environments, or embodiments not seen during training.
- **Few-shot adaptation:** Performance after fine-tuning on a small number of demonstrations.

10.4 Major Datasets

10.4.1 Robot Demonstration Datasets

6 provides a comprehensive list of robot demonstration datasets used for VLA training and evaluation.

10.4.2 Video Datasets for World Model Training

Major video datasets used for world model pre-training include:

- **InternVid** [Wang et al., 2023c]: 7 million video clips with language descriptions.
- **HD-VILA** [Xue et al., 2022]: 100 million video-text pairs from YouTube.
- **WebVid** [Bain et al., 2021]: 10 million video-text pairs for video-language pre-training.
- **Ego4D** [Grauman et al., 2022]: 3,670 hours of egocentric video for first-person interaction understanding.
- **MiraData** [Ju et al., 2024]: A large-scale video dataset for long-form video generation.
- **Gameplay datasets**: Various datasets of game footage used for training neural game engines.

10.4.3 Simulation Datasets

Synthetic data generated in simulation:

- **RoboNet** [Dasari et al., 2020]: 15 million video frames of robot interaction from multiple robot platforms.
- **ORBIT** [Mittal et al., 2023]: A scalable simulation framework for robot learning.
- **Isaac Gym data**: GPU-accelerated simulation data for robot locomotion and manipulation.

10.5 Limitations of Current Evaluation

Despite the progress in benchmarking, several limitations persist:

1. **Benchmark fragmentation:** Different papers evaluate on different benchmarks, making cross-paper comparison difficult. The community would benefit from standardized evaluation suites.
2. **Real-world evaluation cost:** Evaluating VLA models on real robots is time-consuming and expensive, limiting the number of evaluation trials. Simulation benchmarks are more scalable but may not capture real-world challenges.
3. **Metric limitations:** Success rate alone does not capture efficiency, robustness, or generalization quality. More nuanced metrics are needed.
4. **World model evaluation:** Evaluating world models for planning or simulation is more complex than evaluating video generation quality. Standardized protocols for evaluating world model utility for downstream tasks are lacking.
5. **Reproducibility:** Many VLA evaluations are conducted on proprietary hardware or with non-public data, limiting reproducibility. Efforts like OpenVLA and Octo, which provide fully open-source pipelines, are important steps toward addressing this.

10.6 Summary

In this section, we have cataloged the major benchmarks, datasets, and evaluation metrics for VLA models and world models. The evaluation landscape is rapidly evolving, with new benchmarks and datasets being released at an accelerating pace. Standardization of evaluation protocols and increased emphasis on real-world evaluation remain important priorities for the community.

11 Open Challenges and Future Directions

Despite remarkable progress, the fields of VLA models and world models face significant open challenges that must be addressed before these systems can achieve their full potential. In this section, we identify and discuss the most pressing challenges and promising research directions.

11.1 Data Scarcity and Quality

11.1.1 The Robot Data Bottleneck

The most fundamental challenge for VLA models is the scarcity of high-quality robot demonstration data. While language models benefit from trillions of tokens of internet text and vision models from billions of image-text pairs, robot demonstration datasets typically contain only thousands to millions of episodes. This data bottleneck limits the scale and diversity of VLA training.

Potential solutions include:

1. **Autonomous data collection:** Systems that continuously collect and learn from autonomous robot interaction, reducing reliance on human demonstrations. Approaches include goal-conditioned exploration [Pong et al., 2022], curiosity-driven learning [Pathak et al., 2017], and self-improving loops [Pinto and Gupta, 2016].

2. **Scalable teleoperation:** Developing lower-cost, easier-to-use teleoperation systems that democratize data collection. ALOHA [Zhao et al., 2023a] and UMI [Chi et al., 2024] represent steps in this direction.

3. **Cross-embodiment data aggregation:** Aggregating data from diverse robots through initiatives like Open X-Embodiment [Collaboration et al., 2024], creating a "robot internet" of demonstration data.

4. **Video pre-training:** Leveraging the vast amounts of human manipulation video available online (e.g., YouTube, Ego4D) for pre-training visual representations and dynamics understanding, as demonstrated by GR-2 [Bao et al., 2025].

5. **Synthetic data from world models:** Using world models to generate synthetic robot demonstrations [Yang et al., 2024], potentially creating unlimited training data. The quality and diversity of synthetic data remain open questions.

11.1.2 Data Quality and Consistency

Beyond quantity, the quality and consistency of robot demonstration data significantly affect VLA performance. Key quality challenges include:

- **Multi-demonstrator variability:** Different demonstrators have different styles, speeds, and strategies, creating inconsistency in the training data.

- **Suboptimal demonstrations:** Human demonstrations are often suboptimal (wasteful motions, failed grasps, corrections), and learning from suboptimal data can degrade policy quality.
- **Observation quality:** Camera calibration, lighting conditions, and sensor noise vary across data collection sessions, introducing confounding factors.

11.2 Sim-to-Real Transfer

11.2.1 The Reality Gap

The gap between simulated and real-world environments remains a fundamental challenge for both VLA models and world models. Simulations make simplifying assumptions about physics, materials, and sensor characteristics that do not fully capture the complexity of the real world.

Current approaches to bridging this gap include:

- **Domain randomization:** Randomizing simulation parameters during training to improve robustness to real-world variations [Tobin et al., 2017].
- **Real-to-sim:** Building high-fidelity simulations of specific real-world environments using 3D scanning and reconstruction [Shen et al., 2021].
- **World model simulation:** Using world models to generate photorealistic simulated experiences that more closely match the real world [Yang et al., 2024].
- **Fine-tuning with real data:** Pre-training in simulation and fine-tuning with a small amount of real-world data [Chi et al., 2023].

11.2.2 World Model Fidelity

For world models to serve as practical simulators, they must generate sufficiently realistic environment dynamics. Current limitations include:

- Physical inconsistencies (objects passing through each other, gravity violations);
- Short temporal coherence (degradation after a few seconds of generation);
- Limited controllability (difficulty executing precise actions);
- Lack of compositional understanding (inability to generalize to novel object combinations).

11.3 Long-Horizon Reasoning and Task Composition

11.3.1 Temporal Horizon Limitations

Most VLA models predict short action horizons (typically 10–50 timesteps), limiting their ability to plan and execute long-horizon tasks. Extending the planning horizon introduces several challenges:

- **Error compounding:** Small prediction errors accumulate over long sequences, leading to drift from the intended trajectory.
- **Combinatorial explosion:** The number of possible action sequences grows exponentially with the planning horizon.
- **Memory requirements:** Maintaining context over long sequences requires efficient memory mechanisms.

11.3.2 Hierarchical Task Decomposition

A promising approach to long-horizon tasks is hierarchical decomposition: a high-level planner generates subgoals, and a low-level policy achieves each subgoal. Systems like CogAct [Li et al., 2024b] and SayPlan [Rana et al., 2023] demonstrate this approach, but challenges remain in:

- Learning meaningful and achievable subgoal representations;
- Ensuring that the high-level planner and low-level policy are well-coordinated;
- Handling subgoal failures and replanning.

11.3.3 Goal Specification and Language Understanding

Natural language instructions for embodied tasks are often ambiguous, underspecified, or require world knowledge to interpret. For example, "clean the kitchen" requires understanding what constitutes a clean kitchen, what actions are needed, and in what order they should be performed. VLA models must develop robust language understanding that goes beyond simple object-command mappings.

11.4 Safety, Robustness, and Reliability

11.4.1 Safety-Critical Deployment

Deploying VLA models in safety-critical settings (medical, industrial, household) requires:

- **Formal safety guarantees:** Provably ensuring that the policy will not execute dangerous actions. This is challenging for neural network policies that lack formal verification methods.
- **Out-of-distribution detection:** Identifying when the agent encounters situations outside its training distribution, triggering safe fallback behaviors.
- **Robustness to adversarial inputs:** Ensuring that the policy is not easily fooled by adversarial observations or instructions.
- **Graceful degradation:** Failing safely (e.g., stopping) rather than catastrophically when the model's confidence is low.

11.4.2 Uncertainty Quantification

For both VLA models and world models, reliable uncertainty estimation is critical for:

- Detecting when the model is uncertain about its predictions;
- Deciding when to seek human assistance;
- Enabling risk-aware planning that avoids actions with uncertain outcomes.

Current approaches include ensemble methods, Bayesian neural networks, and conformal prediction, but integrating uncertainty estimation into large VLA models remains challenging due to computational cost.

11.5 Computational Efficiency

11.5.1 Real-Time Inference

Real-world robot control typically requires 10–50 Hz control rates, meaning the VLA model must generate actions within 20–100 milliseconds. This is challenging for billion-parameter models, particularly diffusion-based or flow-matching policies that require multiple forward passes per action.

Approaches to improving inference efficiency include:

- Model distillation (training smaller, faster student models);
- Quantization (reducing numerical precision);
- Action chunking (generating multiple actions per inference step);
- Caching (reusing intermediate computations across timesteps);
- Specialized hardware (edge GPUs, TPUs, neural accelerators).

11.5.2 Training Efficiency

Training VLA and world models requires significant computational resources. RT-2 [Brohan et al., 2023a] was trained on TPU pods, and GR-2 [Bao et al., 2025] required training on millions of video clips. Reducing training costs through:

- Parameter-efficient fine-tuning (LoRA, adapters);
- Curriculum learning (training on easy tasks first);
- Mixed-precision training;
- Distributed training optimizations.

11.6 Generalization and Robustness

11.6.1 Zero-Shot and Few-Shot Generalization

The ultimate goal of VLA research is a "universal" robot policy that can perform any task specified in natural language without task-specific training. Current systems demonstrate impressive but limited generalization:

- **Object generalization:** Handling novel objects (well-studied; good progress).
- **Instruction generalization:** Following novel instruction phrasings (moderate progress).
- **Environment generalization:** Operating in novel environments (limited progress).
- **Task generalization:** Performing novel tasks never seen during training (limited progress).
- **Embodiment generalization:** Transferring to novel robot platforms (emerging; see HPT [Allshire et al., 2024]).

11.6.2 Robustness to Distribution Shift

Real-world deployment introduces distribution shifts that can degrade VLA performance:

- Changes in lighting, camera angle, or background;
- Objects with novel appearances, materials, or geometries;
- Dynamic environments with moving people or objects;
- Wear and tear on robot hardware affecting kinematics.

11.7 Toward Unified Foundation Models

11.7.1 The Convergence Hypothesis

A compelling hypothesis is that VLA models and world models will converge toward a single unified foundation model that jointly perceives, predicts, and acts. This "world-action model" would:

1. Process visual observations and language instructions (perception);
2. Predict future states conditioned on candidate actions (world modeling);
3. Select and execute actions to achieve the specified goal (action generation);
4. Learn from both passive video data and active interaction data.

Early evidence for this convergence comes from:

- UniVLA [Li et al., 2025]: Jointly training video generation and action prediction;
- GR-2 [Bao et al., 2025]: Using video prediction pre-training for VLA;
- π_0 [Black et al., 2024]: Demonstrating that world understanding from VLM pre-training enables general manipulation;
- Sora [OpenAI, 2024]: Exhibiting world model properties as a video generation model.

11.7.2 Scaling Laws for Embodied Foundation Models

An important open question is whether the scaling laws observed in language modeling [Kaplan et al., 2020] and vision modeling [Zhai et al., 2022] extend to embodied AI. Preliminary evidence suggests:

- VLA performance improves with model scale (RT-2: 55B vs. RT-1: 35M);
- VLA performance improves with data diversity (RT-X: cross-embodiment > single-robot);
- World model quality improves with scale (Cosmos: 50B+ > smaller models);
- But the relationship may be non-monotonic for certain embodied capabilities (e.g., precise manipulation may not benefit from scale alone).

Understanding the scaling laws for embodied foundation models is a critical research direction that will inform resource allocation and model design decisions.

11.8 Ethical and Societal Considerations

11.8.1 Responsible Deployment

As VLA models become capable of operating in human environments, responsible deployment considerations become paramount:

- **Transparency:** Understanding and explaining why a VLA model took a particular action.
- **Bias and fairness:** Ensuring that VLA models do not exhibit biases in how they interact with different people, objects, or environments.
- **Privacy:** VLA models process visual data from homes and workplaces, raising privacy concerns.
- **Economic impact:** Automation of physical tasks may have significant workforce implications.

11.8.2 Dual-Use Concerns

World models capable of generating realistic interactive environments could potentially be misused (e.g., creating deceptive content, generating deepfakes). Responsible development requires:

- Watermarking and provenance tracking for generated content;
- Safety filters that prevent generation of harmful content;
- Access controls for powerful world model capabilities;
- Ethical review processes for world model research.

11.9 Summary

The challenges facing VLA models and world models are significant but not insurmountable. The key research directions are: (1) addressing the data bottleneck through scalable collection, aggregation, and synthetic generation; (2) improving world model fidelity for practical simulation and planning; (3) extending temporal horizons for long-horizon task execution; (4) ensuring safety and reliability for real-world deployment; (5) improving computational efficiency for real-time control; (6) advancing generalization to novel tasks, environments, and embodiments; and (7) developing unified foundation models that integrate perception, prediction, and action. The convergence of VLA and world model paradigms represents a particularly promising direction toward achieving truly general-purpose embodied intelligence.

12 Conclusion

This survey has provided a comprehensive review of Vision-Language-Action (VLA) models and world models for embodied AI, two rapidly evolving paradigms that are reshaping the landscape of artificial intelligence for physical and virtual agents. We have traced the evolution from classical model-based reinforcement learning and behavioral cloning to the current state of the art, where billion-parameter foundation models learn to perceive, reason, and act from internet-scale data.

12.1 Key Takeaways

Our analysis yields several key insights:

1. **The pre-training paradigm has been successfully extended to embodied AI.** The transition from training robot policies from scratch (RT-1) to fine-tuning pre-trained vision-language models (RT-2, OpenVLA, π_0) has been transformative, enabling VLA models to leverage the semantic knowledge, visual understanding, and reasoning capabilities acquired during internet-scale pre-training. This mirrors the pre-training revolution in NLP and computer vision, suggesting that the same principles apply to embodied control.

2. **Action representation matters as much as model architecture.** The evolution from discretized action tokens (RT-1/RT-2) through diffusion-based policies (Diffusion Policy, Octo) to flow-matching actions (π_0) demonstrates that the choice of action representation is a critical design decision. Continuous action representations (diffusion, flow matching) outperform discretized tokens for tasks requiring precise control and handling multimodal action distributions.

3. **Cross-embodiment training is both feasible and beneficial.** The Open X-Embodiment initiative has demonstrated that a single VLA model can be trained on data from diverse robots and transfer positively to each individual robot. This "robot internet" approach suggests that scale and diversity in robot data are as important as scale in model parameters.

4. **World models have evolved from RL tools to general-purpose simulators.** The progression from latent-space dynamics models (Dreamer family) to observation-space generative simulators (UniSim, Genie, Cosmos) represents a paradigm shift in what world models can do. Modern world models are not just tools for planning within RL—they are interactive simulators that can generate realistic environments for training, testing, and evaluation.

5. **VLA models and world models are converging.** The most promising systems (GR-2, UniVLA, π_0) integrate elements of both paradigms: world model pre-training enhances VLA performance, and VLA-style action conditioning improves world model controllability. This convergence points toward unified foundation models that jointly perceive, predict, and act.

6. **Virtual and real-world applications are mutually reinforcing.** Game environments (Minecraft, Atari, Crafter) provide scalable testbeds for developing embodied AI techniques, while real-world robotics provides the ultimate test of practical capability. Neural game engines (GameNGen, Oasis) represent a direct application of world models that also advances the broader goal of interactive world simulation.

12.2 The Road Ahead

The field of embodied AI is at an inflection point. The convergence of large-scale pre-training, generative modeling, and robot learning is producing systems with capabilities that were unimaginable just a few years ago. We identify several milestones that, if achieved, would represent transformative advances:

- **Near-term (1–2 years):** VLA models that can be deployed in unstructured home environments for a wide range of manipulation tasks, requiring only tens of demonstrations for adaptation. World models that can simulate multi-minute interactive scenarios with physical consistency.

- **Medium-term (3–5 years):** Universal robot policies that control diverse embodiments (arms, hands, mobile manipulators, humanoids) and perform tasks specified in natural language without task-specific training. World models that serve as high-fidelity digital twins of real environments.
- **Long-term (5–10 years):** Unified perception-prediction-action foundation models that learn from the totality of human knowledge (text, images, video, physical interaction) and exhibit general-purpose embodied intelligence across domains. These models would seamlessly integrate the capabilities currently distributed across VLA models, world models, LLMs, and VLMs into a single architecture.

Achieving these milestones will require addressing the challenges outlined in 11: the data bottleneck, sim-to-real transfer, long-horizon reasoning, safety, computational efficiency, and generalization. It will also require continued collaboration across research communities—robotics, computer vision, NLP, reinforcement learning, and game AI—as the boundaries between these fields continue to blur.

12.3 Final Remarks

The vision of machines that can perceive, understand, and act in the physical world has driven AI research since its inception. The systems surveyed in this paper—from RT-1 to $\pi_{0.5}$, from Ha and Schmidhuber's world model to Genie 2 and Cosmos—represent concrete steps toward realizing this vision. While significant challenges remain, the pace of progress suggests that general-purpose embodied AI may be achievable within the coming decade. We hope this survey serves as a useful guide for researchers and practitioners contributing to this exciting frontier.

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